WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – I : INTRODUCTION TO HYDROLOGY

1. Engineering Hydrology & Its Applications

> Hydrology is the science that deals with the occurrence, circulation, distribution and properties of water of the earth and its atmosphere. Engineering hydrology = the part concerned with quantity of water for design, planning and operation of water-resource projects (how much, when, where, how often).

❖Applications

- Design of dams, reservoirs and spillways – design flood, reservoir capacity.
- Design of bridges, culverts, drains – waterway area from peak discharge.
- Irrigation planning – crop water requirement, canal capacity.
- Flood control & forecasting, flood-plain zoning, urban drainage.
- Water supply, hydropower, navigation, ground-water development, drought studies.
- Estimation of yield of catchment, sediment, evaporation losses.

❖Branches / terms

- Catchment/ watershed/ basin: area draining to a common outlet. Precipitation, runoff, stream flow, peak flow, base flow.

2. Hydrologic Cycle

> The continuous circulation of water between earth, atmosphere and oceans driven by solar energy and gravity.

- Evaporation (oceans/ lakes/ soil) + transpiration (plants) → water vapour → condensation (clouds) → precipitation → interception, depression storage, infiltration, percolation → ground water, surface runoff → streams → ocean.
- Total earth water ≈ 1.4×10^9 km^3; 97% salt water; 2.5% fresh (mostly ice); ≈ 0.01% usable.

Water balance (hydrologic budget): $P - R - G - E - T = \Delta S$ i.e. Inflow – Outflow = Change in storage

- P = precipitation, R = surface runoff, G = ground-water outflow, E = evaporation, T = transpiration, ΔS = storage change.
- **Horton's engineering representation:** atmospheric, surface, sub-surface phases.

3. Precipitation

Precipitation = all forms of water (liquid or solid) falling from clouds to earth – **rain, snow, hail, sleet, drizzle, glaze, dew/ frost** (the last two are not generally counted).

❖Conditions for precipitation

- Lifting of air (cooling) → saturation → condensation on **nuclei** → drops grow (collision–coalescence, ice crystal process) → fall.

❖Types (by mechanism of lifting)

- **Convective:** warm moist air rises by surface heating, cools, condenses – **thunderstorms**, short, intense, local (tropical).
- **Orographic:** air forced up **mountains** – heavy on windward side (Western Ghats, Himalayas), rain-shadow on lee side.
- **Cyclonic:** low-pressure systems – **frontal** (warm/ cold front) and **non-frontal** (tropical cyclones, monsoon depressions – A.P. & Odisha coast).
- **Monsoon:** seasonal (SW monsoon June–Sept gives ~ 75% of India's rain; NE monsoon Oct–Dec for Tamil Nadu & coastal A.P.).

❖Forms

- **Rain** (> 0.5 mm drops), **drizzle** (< 0.5 mm), **snow**, **sleet** (frozen rain), **hail** (ice balls from cumulonimbus), **glaze**, **dew**, **frost**, **fog drip**.

❖Intensity classification

Class	Rate (mm/h)
Light	< 2.5
Moderate	2.5 – 7.5
Heavy	> 7.5

4. Rainfall Measurement – Rain Gauges

> Rain gauge = instrument measuring rainfall depth (mm) collected over a standard horizontal area (funnel). Depth = volume/ area.

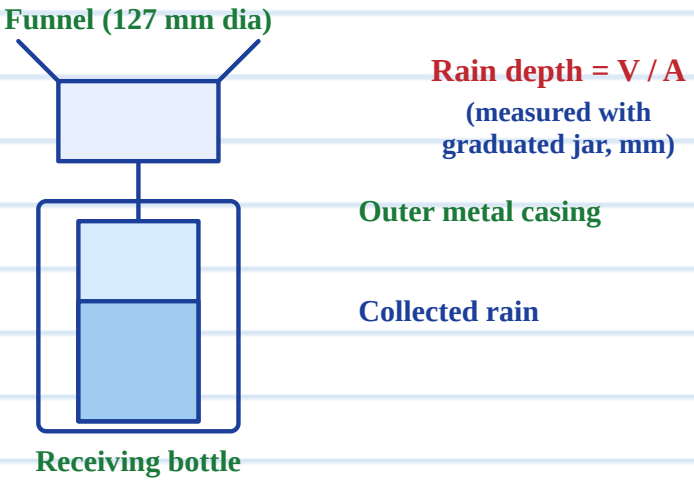


Fig 1.1 Symon's (non-recording) rain gauge

❖A. Non-recording gauges

- **Symon's gauge** (standard in India): **127 mm (5 in.) dia funnel**, bottle, outer casing; read **daily at 08:30 IST** with a graduated measuring jar.
- Limitation: gives only total rainfall; no intensity.

❖B. Recording gauges (give continuous record – hyetograph)

- **Tipping-bucket:** funnel → two small buckets on a pivot; each 0.25/ 0.5 mm fill tips & records electrically. Good for telemetry.
- **Weighing-bucket:** weight of collected rain/ snow recorded by pen on a drum; best for snow.
- **Float (natural syphon / Dines tilting-siphon):** float rises with water, pen traces mass curve on a clock-driven chart; when full the **siphon** empties it automatically – used by IMD.
- **Telemetering / automatic weather stations (AWS)**, radar & satellite rainfall estimates.

❖Siting of a rain gauge (important)

- Open, level ground, away from trees/ buildings – **distance ≥ 2 × height of obstruction**; funnel rim horizontal, **30 cm above ground** (to avoid splash).
- **Density of gauges (IS 4987 / WMO):** plains 1 per 520 km^2; hilly 1 per 260 km^2; heavy-rain hills 1 per 130 km^2; at least 10% recording.
- Wind-shield, error due to wind eddies, evaporation, splash, wetting loss.

5. Presentation & Interpretation of Rainfall Data

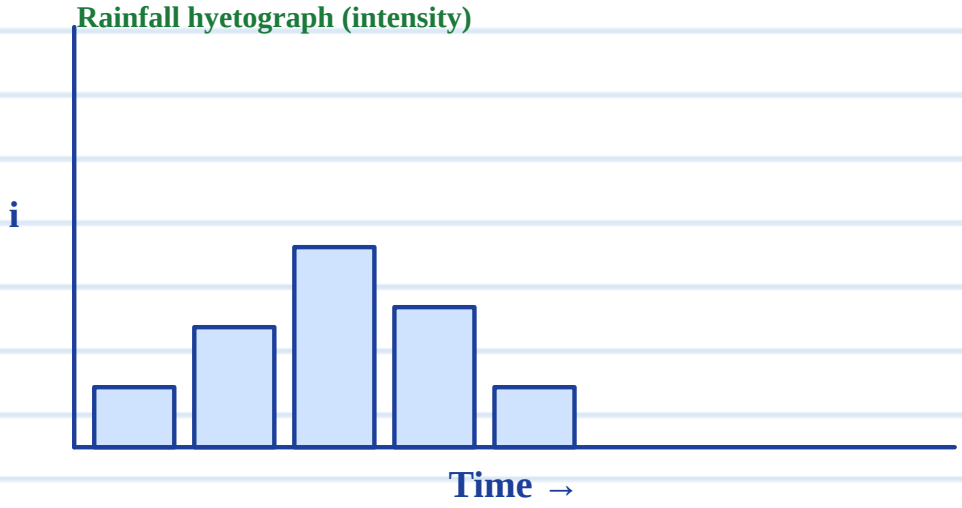


Fig 1.3 Hyetograph – rainfall intensity versus time

- **Hyetograph:** bar graph of rainfall intensity (mm/h) vs time – shows storm pattern.
- **Mass curve of rainfall:** cumulative rainfall vs time – slope = intensity; from recording gauge.
- **Point rainfall** table; **Isohyets** (lines of equal rainfall) maps.
- **Depth–Area–Duration (DAD) curves:** average depth reduces as area increases; used for design storms.
- **Intensity–Duration–Frequency (IDF) curves:** $i = K T^x/(t + a)^n$.
- **Recurrence interval (return period)** $T = (N+1)/m$ (**Weibull**; m = rank in descending order); probability of exceedance $P = 1/T$; risk of occurrence at least once in n years = $1 - (1 - 1/T)^n$.

❖Missing rainfall data

- Station X record missing; nearby stations A, B, C with normal annual rainfalls N_A, N_B, N_C and storm rainfalls P_A, P_B, P_C:

$$P_X = (N_X/n)\,[P_A/N_A + P_B/N_B + P_C/N_C]$$ (if normals differ > 10%: normal-ratio method; else simple arithmetic mean)

❖Consistency of record – double mass curve

- Plot **cumulative annual rainfall of station X vs cumulative average of surrounding stations**; change in **slope** indicates change in gauge conditions/ location. Correct: $P_c = P_a \times (M_c/M_a)$ (slopes).
- **Test of adequacy of gauges:** $N = (C_v/\varepsilon)^2$; $C_v = 100\,\sigma/\bar{P}$; ε = permissible error (10%).

6. Average Rainfall over a Basin

❖(a) Arithmetic mean method

$$\bar{P} = (P_1 + P_2 + \dots + P_n) / n$$

- Simple; suitable for flat area with uniformly distributed gauges & small variation (< 10%).

❖(b) Thiessen polygon method

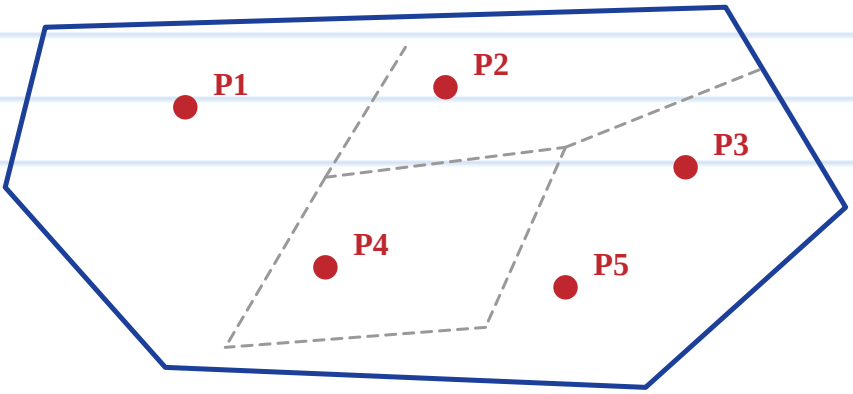


Fig 1.2 Thiessen polygon method (area-weighted rainfall)

- Join adjacent gauges; draw **perpendicular bisectors** → polygons; area A_i of each polygon around gauge i.

$$\bar{P} = \Sigma (P_i A_i) / \Sigma A_i$$

- Accounts for non-uniform gauge distribution; **more accurate than arithmetic**; doesn't consider orography; polygons change if a gauge is added/ missing.

❖(c) Isohyetal method

- Draw **isohyets** (equal rainfall contours) by interpolation; area A_j between successive isohyets; mean rainfall of each strip $P_j = (P_a + P_b)/2$.

$$\bar{P} = \Sigma (P_j A_j) / \Sigma A_j$$

- **Most accurate** (esp. in hills with orographic effects) but needs judgement and many gauges.

❖Comparison

Method	Suitable for	Accuracy
Arithmetic	Flat, uniform gauges	Low
Thiessen	Moderately varied terrain	Medium
Isohyetal	Hilly/ large area	Highest

❖Example

> Four gauges with rainfall (mm) & Thiessen areas (km^2): 80/ 200, 100/ 300, 120/ 400, 60/ 100. $\Sigma P A = 16000 + 30000 + 48000 + 6000 = 100{,}000$; $\Sigma A = 1000 \rightarrow \bar{P} = 100$ mm. Arithmetic mean = (80+100+120+60)/4 = 90 mm.

7. Evaporation

> **Evaporation** = the process by which **liquid water changes to vapour** at a free surface (water, wet soil) below boiling point; needs **latent heat** (≈ 2.45 MJ/kg at 20 °C). **Evapotranspiration (ET)** = evaporation + plant transpiration.

- **Importance:** reservoir losses (India: 1–3 m/yr from reservoirs!), irrigation scheduling, water balance.

❖Factors affecting evaporation

- **Vapour pressure difference** ($e_w - e_a$): e_w = saturated vapour pressure at water surface temperature; e_a = actual vapour pressure of air – **driving force** (Dalton's law).
- **Temperature** (air & water) – higher $T \rightarrow$ higher $e_w \rightarrow$ more evaporation.
- **Wind speed** – removes saturated air layer; E increases with wind up to a limit.
- **Relative humidity** – lower RH, more evaporation.
- **Atmospheric pressure** – lower pressure (altitude) $\rightarrow$ more evaporation.
- **Quality of water** – dissolved salts reduce vapour pressure (sea water ≈ 2–3% less than fresh).
- **Size & depth of water body**, solar radiation, heat storage; **monolayers** (cetyl alcohol) reduce loss.

❖Equations

Dalton: $E = C\,(e_w - e_a)$ (E in mm/day, e in mm Hg)

Meyer's formula: $E_L = K_M\,(e_w - e_a)\,(1 + u_9/16)$ mm/day

- u_9 = wind speed (km/h) at 9 m above ground; K_M = 0.36 (large deep waters), 0.50 (small shallow waters).
- Other methods: water-budget, energy-budget, Penman, mass-transfer, empirical (Rohwer).

❖Measurement of evaporation

- Evaporation pans:
  - Class A land pan (US Weather Bureau): 1210 mm dia × 255 mm deep, galvanised iron, on wooden platform; water depth 18–20 cm; hook gauge reading.
  - ISI (Indian Standard) pan: 1220 mm dia × 255 mm, with a fine-wire mesh cover; pan coefficient 0.8.
  - Colorado sunken pan, floating pan (best for lakes).
- Lake evaporation = pan coefficient × pan evaporation: $E_L = C_p\,E_p$; $C_p \approx 0.70$ (Class A), 0.8 (ISI), 0.78 (Colorado), 0.80 (floating).
- Atmometers/ evaporimeters (Piche) for lab; lysimeters for ET.
- Control of evaporation: reduce exposed area (deep narrow reservoirs), wind-breaks, monomolecular films (hexadecanol), storing water underground, spray shading.

❖Example

> Class A pan evaporation = 8 mm/day, C_p = 0.7 → lake evaporation = 5.6 mm/day.

> Meyer: e_w = 20 mm Hg, e_a = 10 mm Hg, u_9 = 16 km/h, K_M = 0.36: E = 0.36 × 10 × (1 + 1) = 7.2 mm/day.

8. Infiltration

> Infiltration = movement of water from the ground surface into the soil. Infiltration rate f (mm/h) = depth entering per unit time; infiltration capacity f_p = maximum rate under given conditions. Percolation = downward movement within soil to water table.

❖Factors affecting infiltration

- Soil type & texture (sand high, clay low), soil structure/ porosity, initial moisture content (dry soil – high initial rate).
- Surface cover/ vegetation (roots & litter ↑), land use, compaction/ crusting (rain-drop sealing ↓), slope.
- Rainfall intensity (rate cannot exceed supply), depth of ponding, temperature/ viscosity, entrapped air, quality of water (silt, salts), tillage.

❖Horton's infiltration equation

$$f_p = f_c + (f_0 - f_c)\, e^{-kt}$$

- f_0 = initial capacity, f_c = final (constant) capacity, k = decay constant (h^{-1}); rate decreases exponentially with time as soil fills.
- Cumulative infiltration $F(t) = f_c\, t + (f_0 - f_c)(1 - e^{-kt})/k$. Other models: Green–Ampt, Philip, Kostiakov.

❖Measurement of infiltration

- Double-ring infiltrometer: two concentric rings (30 cm inner, 60 cm outer) driven in soil; water maintained at constant depth; outer ring prevents lateral spreading; inner ring reading of water added at intervals gives f.
- Single-ring infiltrometer (lateral spread error); rainfall simulators (sprinkler plots); hydrograph analysis; tension infiltrometer.

❖Infiltration indices (avg losses over a storm)

- Φ-index: the constant rate of rainfall above which the volume of rainfall = volume of direct runoff. Rain below Φ is assumed all infiltration. Find by trial: $\Sigma\ (i - \Phi)\,\Delta t = R$ (runoff depth) for intervals with $i > \Phi$.
- W-index: average infiltration rate = $(P - R - S_d)/t_r$, corrected for depression storage S_d and rainfall duration where $i > f$. $W > \Phi$ (since W removes initial losses).

- **Φ-min index** for wet conditions.

❖Example (Φ-index)

> Hourly rainfall (cm): 1.0, 2.0, 3.0, 2.0, 1.0; runoff depth $R = 4.0$ cm. Assume $\Phi = 1.5$: $\Sigma(i - 1.5)$ for $i > \Phi = (2-1.5)+(3-1.5)+(2-1.5) = 2.5$ (but 1.0 below) ✗. Try $\Phi = 1.0$: $(1+2+1) = 4.0$ ✔ ($i = 1.0 \rightarrow 0$). Hence $\Phi = 1.0$ cm/h.

9. Runoff

> **Runoff** = that part of precipitation that appears in streams/ rivers as surface flow. **Total runoff** = surface runoff + interflow + base flow (ground-water flow). **Direct runoff** = surface + prompt interflow.

❖Factors affecting runoff

- **Climatic:** rainfall **intensity, duration, areal distribution,** storm direction, antecedent rainfall, evaporation, temperature, season.
- **Catchment (physiographic): area, shape** (fan-shaped gives high peak vs elongated), **slope, elevation, orientation, soil type & geology,** land use & vegetation, **drainage density,** depression storage, lakes/ ponds, urbanisation (impervious cover ↑ runoff).
- **Hydraulic:** channel storage, stream density, infiltration.

❖Computation of runoff (volume / yield)

- **Strange's table** (rainfall, soil class, yield %), **Binnie's percentages, Barlow's coefficients** (runoff coefficient from table; $Q = K_b P$), **Khosla's formula** (temperature based):

> Khosla's formula: $R = P - L$ (L = evaporation/ infiltration loss, estimated from mean temperature T of the catchment)

- **Inglis & De Souza formula (Maharashtra):** $R = 0.85P - 30.5$ (Ghat regions) ; $R = (1/254)\,P\,(P - 17.8)$ (plains) (R, P in cm).
- **SCS-CN (Curve Number) method (USDA):**

> $Q = (P - I_a)^2 / (P - I_a + S)$, $I_a = 0.2S \rightarrow Q = (P - 0.2S)^2 / (P + 0.8S)$; $S = 25400/CN - 254$ (mm)

- CN = 100 (impervious) ... 30; depends on soil group A–D, land use, antecedent moisture condition (AMC I, II, III); valid when $P > 0.2S$.

❖Example (SCS)

> $P = 100$ mm, $CN = 80$: $S = 25400/80 - 254 = 63.5$ mm; $I_a = 12.7$; $Q = (100 - 12.7)^2/(100 + 50.8) = 7621/150.8 = 50.5$ mm (runoff coefficient ≈ 0.5).

❖Runoff coefficient

C = Runoff / Rainfall ; typical C: roofs 0.9; pavements 0.8; sandy lawns 0.1–0.2; clay 0.5; forests 0.1–0.2; cultivated 0.3

10. Design Flood & Maximum Rate of Run-off

> Design flood = the flood adopted for the design of a hydraulic structure (spillway, bridge, weir) corresponding to a safe, economical risk (return period). Types: Probable Maximum Flood (PMF), Standard Project Flood (SPF), Maximum Probable Flood (MPF); frequency flood (e.g., 100-yr, 1000-yr).

- Return period (IS 11223): large dams (PMF), small structures (25–100 yr), bridges (50–100 yr), culverts (25 yr), urban drains (2–10 yr).

❖Estimation of maximum rate of runoff (peak discharge)

(a) Empirical formulas (Q_{max} in m^3/s, A in km^2)

- Dicken's: $Q = C_D A^{3/4}$ (C_D = 11.4 for N. India hills, 13.9–19.5 central, 22–25 W. Ghats, A in km^2).
- Ryves: $Q = C_R A^{2/3}$ (C_R = 6.8 near coast (80 km), 8.5 (80–160 km), 10.1 (limited hills)).
- Inglis: $Q = 124 A / \sqrt{(A + 10.4)}$ (Western Ghats – Maharashtra).
- Nawab Jung Bahadur (Hyderabad/ A.P.): $Q = C_N A^{(0.993 - 1/14 \log A)}$, C_N = 48 – 60.
- Limitations: region-specific, ignore storm characteristics.

(b) Rational method (small catchments < 50 km^2)

$Q_p = C i A / 360$ (Q in m^3/s, i in mm/h, A in hectares) ≡ $Q = 0.278 C i A$ (A in km^2)

- i = rainfall intensity for duration = time of concentration t_c (time for water from the most remote point to reach outlet). Kirpich: $t_c = 0.0195\,L^{0.77}\,S^{-0.385}$ (min; L in m, S slope m/m).
- Assumptions: uniform intensity over whole area for duration t_c, C constant.

(c) Unit hydrograph method, (d) Flood-frequency analysis (Gumbel, log-Pearson III), (e) Envelope curves.

❖Gumbel's method (flood frequency)

$$x_T = \bar{x} + K\sigma\,,\quad K = -(\sqrt{6}/\pi)\,[\,0.5772 + \ln\ln(\,T/(T-1)\,)\,]$$

❖Examples

Dicken's: $A = 100\ km^2$, $C_D = 14 \rightarrow Q = 14 \times 100^{0.75} = 14 \times 31.6 = 443\ m^3/s$.

Rational: $A = 200$ ha, $C = 0.5$, $i = 36$ mm/h $\rightarrow Q = 0.5\times36\times200/360 = 10\ m^3/s$.

Inglis: $A = 50\ km^2 \rightarrow Q = 124\times50/\sqrt{60.4} = 798\ m^3/s$.

II. Separation of Base Flow

Hydrograph separation = splitting the stream flow hydrograph into direct runoff (DRH) and base flow (ground-water contribution); needed to derive unit hydrographs.

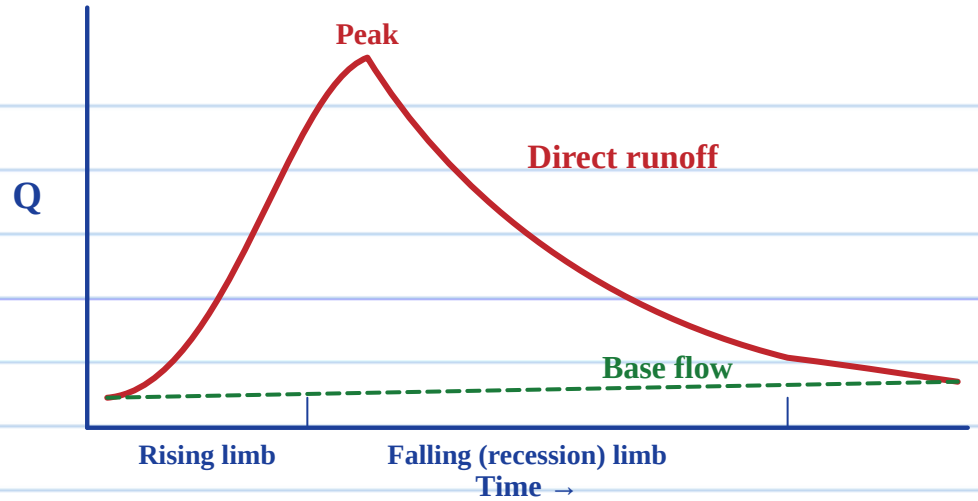


Fig 1.4 Hydrograph with base-flow separation (straight-line method)

❖Methods

- **Straight-line method (Method I):** join the start of rising limb (A) to the point on recession limb (B) where direct runoff ends, by a horizontal/ straight line; area above = DRH. Simple.
- **Fixed base length:** end of direct runoff after N days = $0.827\,A^{0.2}$ (A in km²); line from A (start) extended; the recession curve extended backward from point of inflection.
- **Variable slope method:** project the pre-storm recession curve to the peak time, join to the point N days after the peak.
- **Recession (master) curve analysis:** $Q_t = Q_0\,K_r^t$ (K_r = recession constant 0.93–0.95 for storage); plot **log Q vs t**; break in slope shows end of surface runoff.
- **Direct-runoff depth** = volume of DRH / catchment area.

$$\text{Direct runoff depth (cm)} = [\Sigma\ \text{DRH ordinates } (m^3/s) \times \Delta t\ (s)] / [A\ (m^2)] \times 100$$

❖Example

Catchment 36 km². DRH ordinates sum 90 m³/s at 6-h intervals → volume = $90\times6\times3600 = 1.944\times10^6$ m³. Depth = $1.944\times10^6/36\times10^6 = 0.054$ m = **5.4 cm.**

12. Important Questions – Unit I

- Define hydrology; applications. Explain the hydrologic cycle with a sketch.
- Types and forms of precipitation. Describe recording & non-recording rain gauges.
- Methods of computing average rainfall (Thiessen, isohyetal); double mass curve; missing data.
- Factors affecting evaporation; measurement; Meyer's formula; control.
- Infiltration – factors, measurement (double ring), Horton equation, Φ & W indices.
- Factors affecting runoff; SCS-CN; empirical formulas; rational method.
- Design flood; separation of base flow.

Formula sheet: $\bar{P} = \Sigma PA/\Sigma A$; $Q = CiA/360$; $S = 25400/CN - 254$; $f = fc + (f0 - fc)e^{(-kt)}$; $N = 0.827A^{0.2}$

WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – II : HYDROGRAPH ANALYSIS & GROUND WATER

I. Hydrograph

> A hydrograph is a graph of discharge (Q) versus time (t) at a given section of a stream. It integrates all catchment characteristics and the storm that caused the flow.

- Types: annual hydrograph (daily/ monthly flows – perennial or intermittent streams), flood (storm) hydrograph (single storm), hourly hydrograph; unit hydrograph; synthetic hydrograph.

❖Components of a flood hydrograph

- Rising limb (concentration curve): discharge increasing; shape depends on storm intensity & catchment, antecedent conditions.
- Crest segment (peak): maximum discharge Q_p, occurs after time to peak t_p.
- Falling/ recession limb: decreasing flow after rain ends; controlled by catchment storage (depletion curve).
- Base flow: ground-water contribution. Direct runoff = surface runoff + interflow.
- Basin lag (t_l): time from centroid of rainfall to peak of hydrograph. Time base T_b: total duration of direct runoff. Time of concentration t_c.

❖Factors affecting shape

- Catchment area, shape (fan vs elongated), slope, drainage density, soil/ land use, storm intensity, duration, movement; urbanisation → higher & earlier peak.

❖Recession equation

$$Q_t = Q_0 K_r^t \text{ or } Q_t = Q_0 e^{-at} \quad (K_r = \text{recession constant} < 1)$$

2. Unit Hydrograph (UH) – Sherman 1932

> The unit hydrograph of a catchment is the direct-runoff hydrograph resulting from 1 cm (unit depth) of net (effective) rainfall, uniformly distributed over the basin, occurring at a uniform rate for a specified duration D hours. Called D-hour unit hydrograph.

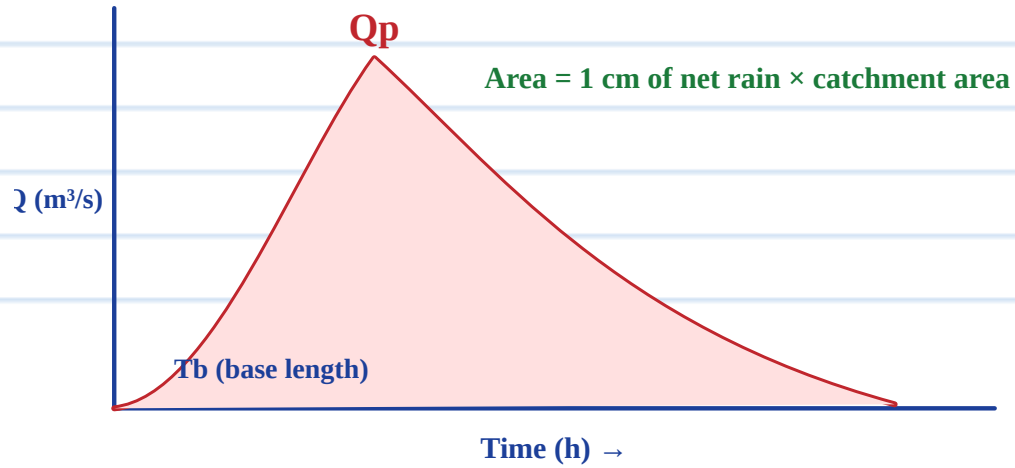


Fig 2.1 Unit hydrograph (UH) of duration D hours

❖Basic assumptions

- 1. Constant (uniform) intensity of effective rainfall for duration D.
- 2. Uniform areal distribution over the entire catchment.
- 3. Time invariance: same hydrograph for same effective rain duration irrespective of when it occurs.
- 4. Linearity (superposition): ordinates of DRH are directly proportional to effective rainfall; hydrographs of successive rains can be added (principle of proportionality & superposition).
- 5. Time base of direct runoff is constant for rains of equal duration.

❖Properties

- Area under UH = volume = 1 cm × A (catchment area).
- Ordinates scale with rain depth: DRH ordinates for r cm = r × UH ordinates.
- Peak flow and time to peak depend on D.

3. Construction (Derivation) of Unit Hydrograph from a Storm Hydrograph

❖Steps

- 1. Select an isolated, simple, uniform-intensity storm of duration D with 1–5 cm effective rain.
- 2. Obtain stream-flow hydrograph; separate base flow (straight line/ fixed base).
- 3. Calculate direct runoff volume; depth of direct runoff = volume/ catchment area (= net rainfall R_e in cm).
- 4. Divide DRH ordinates by R_e → ordinates of D-h UH.
- 5. Check area under UH = 1 cm (volume = A × 0.01 m).
- 6. Plot UH vs time.

❖Example – derivation of 6-h UH

> Catchment A = 36 km^2; total hydrograph & base flow (m^3/s):

t (h)	0	6	12	18	24	30	36
Total flow	10	50	120	90	50	25	10
Base flow	10	10	10	10	10	10	10
DRH	0	40	110	80	40	15	0

$\Sigma DRH = 285\ m^3/s$ → volume = $285 \times 6 \times 3600 = 6.156 \times 10^6\ m^3$; $R_e = 6.156\times10^6/(36\times10^6) = 0.171\ m = 17.1$ cm. UH ordinates = DRH/17.1 → 0, 2.34, 6.43, 4.68, 2.34, 0.88, 0 m^3/s (check area ≈ 1 cm).

4. Application – Flood Hydrograph from Unit-Duration Rainfall

❖Using UH to get DRH for a given storm (convolution)

- Given: D-hour UH ordinates $U_1, U_2, \ldots$; storm with effective rainfall depths $R_1, R_2, \ldots$ in successive D-hour intervals.
- Principle: DRH from each block = R_j × UH, shifted by D hours for each next block; add (superpose) ordinates.

$$Q_n = \Sigma_{m=1}^{n} R_m \cdot U_{n-m+1} \text{ (discrete convolution)}$$

- Then add base flow to get total flood hydrograph.

❖Example

> 2-block storm: effective rain 2 cm (first 6 h), 3 cm (next 6 h). 6-h UH ordinates (at 6-h steps): 0, 5, 20, 15, 8, 3, 0.

- 2 cm hydrograph: 0, 10, 40, 30, 16, 6, 0, 0
- 3 cm hydrograph (lagged 1 step): 0, 0, 15, 60, 45, 24, 9, 0
- DRH = sum: t=0: 0; 6: 10; 12: 55; 18: 90; 24: 61; 30: 30; 36: 9; 42: 0. Peak = 90 m^3/s at 18 h. Add base flow (say 10) → total peak 100 m^3/s.

❖Other uses of UH

- Flood forecasting & design flood (from design storm), reservoir routing inputs, effect of land-use change, synthetic UH for ungauged basins (Snyder's method: $t_p = C_t (L L_{ca})^{0.3}$; SCS triangular UH: $Q_p = 2.08\ A / t_p$, with A in km^2, Q in m^3/s per cm).

❖Limitations of the unit hydrograph

- Applicable only to catchments of 25 – 5000 km^2 (very small: rational; very large: non-uniform rain, channel routing needed).
- Assumes uniform rainfall distribution & intensity, which rarely occurs; non-linear behaviour in large floods & very small ones.
- Requires accurate gauge data; difficult to separate base flow; snow-melt effects & storage reservoirs not handled.
- Catchment change (urbanisation) invalidates; rainfall duration must be near D (D ≈ 1/4 of basin lag).
- Not applicable to areas with large temporary storage or ungauged (needs synthetic UH).

5. S-Hydrograph (S-Curve)

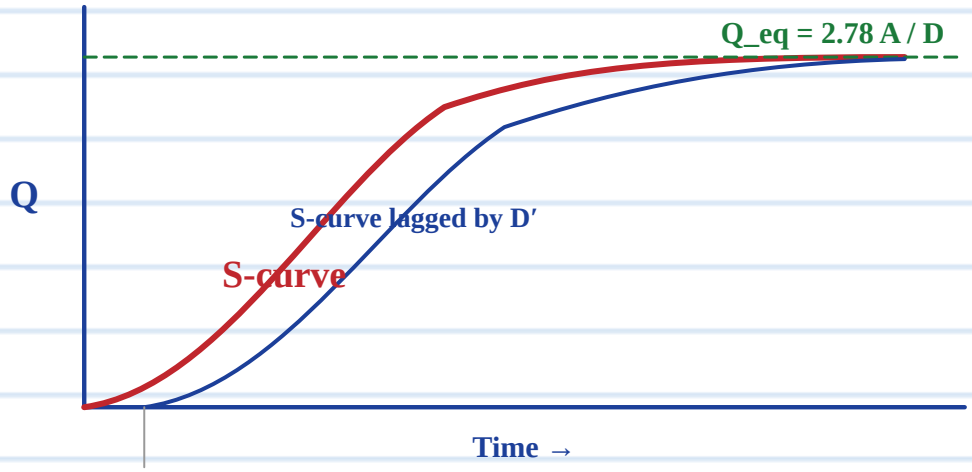


Fig 2.2 S-hydrograph: summation of continuous series of D-h unit hydrographs

S-hydrograph = hydrograph produced by a **continuous effective rainfall at a constant rate of 1 cm/ D hours** (i.e., intensity 1/D cm/h) over an **indefinite period**; obtained by **summing D-hour UHs, each lagged by D hours.**

- Shape: rises like an "S" and **becomes constant at equilibrium discharge Q_e** when inflow = outflow:

$$Q_e = 2.78\ A / D\ (m^3/s;\ A \text{ in } km^2,\ D \text{ in hours})\ [\text{since } Q_e = A \times (1\ cm / D)]$$

❖Construction

- 1. Write the D-h UH ordinates at intervals D. 2. Lag each ordinate series by D, 2D, 3D, ... 3. **Sum** of all → S-curve ordinates S(t) (equilibrium after about T_b/D UHs). 4. Smooth; check Q_e = 2.78 A/D.

❖Use – derive a UH of different duration D'

- Lag the S-curve by D' hours, **subtract** from original: S(t) – S(t – D') = hydrograph for intensity (1/D) acting D' hours = (D'/D) cm of rain.

$$D'\text{-h UH ordinates} = (D / D')\ [S(t) - S(t - D')]$$

- Eg. from 6-h UH get 3-h UH or 12-h UH. Also used when D' is not a multiple of D (not possible by direct lagging).
- If S-curve oscillates (non-uniform data) → hump; smooth it.

❖Example

6-h UH ordinates: 0, 5, 20, 15, 8, 3, 0. S-curve: 0, 5, 25, 40, 48, 51, 51 (equilibrium 51 → $A = 51 \times 6/2.78 = 110\ km^2$). 12-h UH (D' = 12, D = 6): ordinates (S(t) − S(t−12))×(6/12): t=6: 5/2 = 2.5; t=12: (25−0)/2 = 12.5; t=18: (40−5)/2 = 17.5; t=24: (48−25)/2 = 11.5; t=30: (51−40)/2 = 5.5; t=36: (51−48)/2 = 1.5.

6. GROUND WATER – Introduction

Ground water = water **below the ground surface in the saturated zone** (pores/ fissures of soil & rock); ~ 30% of fresh water, ~ 60% of India's irrigation & 85% of rural drinking water. Recharged by **rain infiltration, rivers, canals, tanks, irrigation return flow, artificial recharge.**

- **Advantages:** reliable (less evaporation), local, good quality, drought buffer. **Problems:** over-exploitation (falling water table in A.P., Punjab, Gujarat), fluoride/ arsenic/ salinity, seawater intrusion.
- **Study includes:** occurrence, movement, exploration, well hydraulics, recharge, quality.

7. Aquifer, Aquiclude, Aquifuge & Aquitard



Fig 2.3 Sub-surface water zones and aquifer types

- **Aquifer:** geologic formation that stores and transmits water in sufficient quantity to wells/ springs – sand, gravel, weathered or fractured rock, limestone, sandstone. (Good porosity and permeability.)

- **Aquiclude:** formation that stores water (porous) but does not transmit it appreciably – clay, shale (high porosity, very low permeability).
- **Aquifuge:** formation neither stores nor transmits water – solid granite/ unfractured rock/ compact basalt.
- **Aquitard:** poorly permeable formation (silty clay) that transmits water slowly – allows leakage between aquifers (leaky layer).

❖Summary

Term	Stores water?	Transmits water?	Example
Aquifer	Yes	Yes	Sand, gravel
Aquiclude	Yes	No	Clay
Aquifuge	No	No	Massive granite
Aquitard	Yes	Very slowly	Sandy clay

8. Aquifer Parameters

❖(a) Porosity (n)

Porosity = ratio of volume of voids (pores) to total volume of the soil/ rock mass.

$$n = V_v / V \times 100\ \% \text{ ; relation with void ratio e: } n = e / (1 + e)$$

- Typical: clay 40–55%, silt 35–50%, sand 25–40%, gravel 25–40%, sandstone 5–30%, fractured granite 1–10%.
- **Primary** (original pores) & **secondary** (fractures, solution cavities) porosity. **Effective porosity** = interconnected pores (what yields water).

❖(b) Specific yield (S_y) and (c) Specific retention (S_r)

Specific yield = ratio of the volume of water that drains by gravity from a saturated soil to the total volume of soil. **Specific retention** = ratio of volume of water retained against gravity (by molecular & capillary forces) to total volume.

$$n = S_y + S_r \text{ ; } S_y = V_{drained}/V \text{ ; } S_r = V_{retained}/V$$

Material	n %	S_y %	S_r %
Clay	45	3	42
Silt	45	8	37
Fine sand	35	21	14
Coarse sand	30	27	3
Gravel	25	22	3

- S_y decides the usable storage in unconfined aquifers; coarse materials have high S_y, clay high S_r.
- Volume of water released from an aquifer of area A with water-table fall Δh: $V = S_y A \Delta h$.

❖Example

> Sample of 1000 cm^3 saturated sand yields 250 cm^3 by drainage and retains 100 cm^3: S_y = 25%, S_r = 10%, n = 35%.

> Water table falls 2 m over 5 km^2, S_y = 0.15: $V = 0.15 \times 5 \times 10^6 \times 2 = 1.5 \times 10^6$ m^3.

9. Divisions of Sub-surface Water

> Sub-surface water is divided vertically into two zones by the **water table**:

❖1. Zone of aeration (unsaturated/ vadose zone) – pores partly water, partly air

- (i) **Soil-water (root) zone**: near ground, used by plants; capillary & hygroscopic water.
- (ii) **Intermediate (vadose) zone**: between root zone and capillary fringe; water held by hygroscopic & capillary forces; gravity water passes through.
- (iii) **Capillary fringe (zone)**: just above the water table; water raised by **capillary action** (height $h_c = 0.15/d$ cm; clay ≈ 2–4 m, sand ≈ 0.1–0.5 m).

❖2. Zone of saturation (phreatic zone)

- All pores filled with water; water here is **ground water**; its upper surface = **water table** (pressure = atmospheric).

❖Types of subsurface water

- Hygroscopic, capillary, gravitational (free) water, combined/ chemically bound, juvenile (magmatic), connate (trapped in sediments), meteoric.

10. Water Table

> Water table (phreatic surface) = the upper surface of the zone of saturation, where the water pressure = atmospheric; level observed in an open well.

- Shape: follows the topography (subdued); slopes toward streams/ wells; cone of depression around a pumping well; rises after rains/ recharge, falls in summer & by pumping; fluctuations measured by observation wells/ piezometers.
- Piezometric (potentiometric) surface – imaginary surface to which water rises in a well in a confined aquifer; may be above the ground → artesian (flowing) well.
- Depth of water table: deep (> 8 m) → favourable; shallow (< 2 m) → water-logging.
- Perched water table: local saturated layer above an aquiclude lens, above the main water table.
- Uses: direction of ground-water flow, recharge/ discharge estimate, depth to drill wells.

11. Types of Aquifers

- 1. Unconfined (water-table/ phreatic) aquifer: upper boundary = water table, open to atmosphere; recharged directly from surface; S_y = 0.01 – 0.30; well water stands at water table.
- 2. Confined (artesian/ pressure) aquifer: sandwiched between two impermeable layers (aquicludes); water under pressure > atmospheric; recharge area at outcrop far away; storage coefficient 10^{-5} – 10^{-3}; wells may be flowing artesian.
- 3. Semi-confined (leaky) aquifer: one or both confining layers are aquitards allowing leakage.
- 4. Perched aquifer: small unconfined body held above main water table by a lens of impermeable material.
- 5. Others: alluvial (river sands – high yield), hard-rock (fractured/ weathered granite, basalt) – typical of A.P. & Telangana; coastal (salinity), karstic limestone.

❖Unconfined vs confined

Feature	Unconfined	Confined
Upper boundary	Water table	Impermeable layer
Pressure	Atmospheric	> Atmospheric
Storage coefficient	0.01–0.30 ($\approx S_y$)	10^{-5}–10^{-3}
Water released by	Dewatering pores	Compression of aquifer & expansion of water
Recharge	Local rainfall	Distant outcrop
Well	Normal	Artesian (may flow)

12. Storage Coefficient (Storativity, S)

> Storage coefficient = volume of water an aquifer releases from (or takes into) storage per unit surface area of aquifer per unit change in head (h) normal to that surface (dimensionless).

$$S = \Delta V / (A \cdot \Delta h)$$

- Unconfined: $S \approx S_y$ (0.01–0.30) as water comes from drainage of pores.
- Confined: $S = \rho\, g\, b\, (\alpha + n\beta) \approx 10^{-5} - 10^{-3}$ (b = thickness, α = aquifer compressibility, β = water compressibility) – tiny because water comes from elastic compression.
- Specific storage $S_s = S/b$ (per metre).
- Determined by pumping tests (Theis method) with observation wells.

13. Coefficient of Permeability (Hydraulic Conductivity, K)

> Darcy's law: velocity of flow through a saturated porous medium is proportional to hydraulic gradient: $v = K i$, $i = dh/dL$.

$$Q = K i A = K (\Delta h / L) A \ (m^3/s) ; K = [\text{m/s or m/day}]$$

- Coefficient of permeability K = discharge per unit area per unit hydraulic gradient (velocity under unit gradient) – depends on soil (grain size) and fluid (viscosity, density): $K = k (\rho\, g / \mu)$, k = intrinsic permeability.
- Range: gravel 10^{-1}–10^{1} cm/s; clean sand 10^{-3}–10^{-1}; silty sand 10^{-5}–10^{-3}; silt 10^{-6}–10^{-4}; clay < 10^{-7} cm/s.
- Hazen's formula: $K = C\, d_{10}^{\,2}$ (cm/s, d_{10} in mm; $C \approx 100$–150).
- Seepage velocity $v_s = v/n$ (n = porosity).

- Determination: constant-head test (coarse), falling-head test (fine soil): $K = (aL/At)\ln(h_1/h_2)$, pumping test (field), tracer test.
- Validity of Darcy's law: laminar flow, Reynolds number $Re < 1 - 10$.
- Layered aquifer: equivalent horizontal $K_h = \Sigma K_i b_i / \Sigma b_i$; vertical $K_v = \Sigma b_i / \Sigma (b_i/K_i)$.

14. Transmissibility (Transmissivity, T)

> Transmissibility = rate of flow of water through a vertical strip of aquifer of unit width, extending full saturated thickness, under unit hydraulic gradient.

$$T = K\,b\ (m^2/day)\ ;\ Q = T\,i\,B\ (B = width)$$

- b = saturated thickness (unconfined: $b = h$, varies).
- T high ($> 1000\ m^2/day$) → excellent aquifers (alluvium); 100–1000 good; < 10 poor.

❖Well hydraulics (steady radial flow – Dupuit/ Thiem)

- Unconfined aquifer well: $Q = \pi K (h_2^2 - h_1^2) / \ln(r_2/r_1)$
- Confined aquifer well: $Q = 2\pi K b (h_2 - h_1) / \ln(r_2/r_1) = 2\pi T (h_2 - h_1) / \ln(r_2/r_1)$
- Also in terms of drawdowns s_1, s_2: $T = Q \ln(r_2/r_1) / [2\pi(s_1 - s_2)]$.
- Specific capacity = Q/s (m^3/h per m drawdown); radius of influence R (Sichardt: $R = 3000\, s\sqrt{K}$).
- Theis non-equilibrium: $s = (Q/4\pi T)\,W(u)$, $u = r^2S/(4Tt)$.

❖Solved problems

> P1. Sand aquifer $K = 20$ m/day, thickness $b = 15$ m: $T = 20 \times 15 = 300\ m^2/day$.

> P2. Seepage: $K = 10^{-3}$ cm/s $= 10^{-5}$ m/s, gradient 0.01, area $100\ m^2$: $Q = K\,i\,A = 10^{-5} \times 0.01 \times 100 = 1 \times 10^{-5}\ m^3/s = 0.864\ m^3/day$.

> P3. Confined aquifer $b = 20$ m, $K = 30$ m/day; drawdown heads: $h_1 = 18$ m at $r_1 = 10$ m, $h_2 = 19.5$ m at $r_2 = 100$ m: $Q = 2\pi \times 30 \times 20 \times (1.5)/\ln 10 = 5655/2.303 = 2456\ m^3/day$.

P4. Unconfined well: $K = 40$ m/day, $h_1 = 20$ m ($r = 0.3$ m), $h_2 = 25$ m ($r = 300$ m): $Q = \pi \times 40 \times (625 - 400)/\ln(1000) = 28274/6.908 = \mathbf{4093\ m^3/day}$.

P5. Porosity 0.4, $K = 1$ m/day, $i = 0.005 \rightarrow v = 0.005$ m/day; seepage velocity $= 0.005/0.4 = \mathbf{0.0125}$ m/day.

15. Important Questions – Unit II

- Define unit hydrograph; state assumptions & limitations. Derive a UH from a storm hydrograph.
- Explain construction of S-curve and derivation of UH of different duration; use in flood hydrograph.
- Distinguish aquifer, aquiclude, aquifuge. Porosity, specific yield, specific retention (with relation).
- Divisions of sub-surface water; water table; types of aquifers.
- Define storage coefficient, permeability and transmissibility; Darcy's law; well formulas.

Key relations: $n = Sy + Sr$; $T = K\,b$; $Qe = 2.78\,A/D$; $UH(D') = (D/D')\,[S(t) - S(t-D')]$

WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – III : IRRIGATION & WATER REQUIREMENT OF CROPS

I. Irrigation – Introduction

> Irrigation = the artificial application of water to soil to supply moisture essential for plant growth when rainfall is insufficient or uneven.

- India: net sown area ≈ 140 M ha; net irrigated ≈ 70 M ha (~ 50%); irrigation potential created ≈ 140 M ha; major sources – canals (≈ 25%), tube wells & wells (≈ 60%), tanks.
- Irrigation engineering = planning, design & operation of works (dams, weirs, canals, regulators, drains).

❖Necessity & importance

- Insufficient rainfall (arid/ semi-arid: Rajasthan, Rayalaseema); uneven distribution in time (monsoon only 3–4 months) and place; growing perennial crops (sugar cane, banana) needing water year-round.
- Commercial crops (cotton, rice, sugarcane) need assured water; improves crop yield (irrigated yield 2–3× rain-fed); allows multiple cropping.
- Food security, green revolution, drought protection (famine), rural employment & income, hydropower & navigation from multipurpose projects, recharge of ground water.

❖Advantages of irrigation

- Increase in crop yield & food production; cash crops; protection from famine; increased land value, general prosperity; hydropower generation, drinking water & navigation; ground water recharge; employment; improves climate; afforestation; revenue to government.

❖Ill-effects of irrigation

- Water-logging (rise of water table) & salinity/ alkalinity (soil degradation); climate change locally (humidity ↑), mosquitoes → malaria & water-borne diseases; over-irrigation → seepage losses; dampness in houses; cost of drainage; rise of water table due to canal seepage; submergence & displacement by dams; unequal distribution of water (head vs tail reaches).

2. Types of Irrigation

- Based on source/ scheme:
  - Surface irrigation – gravity flow using canals (perennial from rivers with storage; inundation canals during floods), tanks (A.P. – small tanks), lift irrigation (pumps from rivers/ canals).
  - Ground-water irrigation – open wells, tube wells, bore wells.
  - Rain-water harvesting/ micro-irrigation.
- Based on size (CCA): Major (> 10,000 ha), Medium (2,000–10,000 ha), Minor (< 2,000 ha) – e.g., Nagarjuna Sagar, Srisailam (major), tanks (minor).
- Based on nature of supply: Perennial irrigation (canal water throughout the year – storage reservoir/ weir with assured flow) and Inundation/ flood irrigation (river flood water drawn without head works) & non-perennial.
- Based on purpose: productive (revenue), protective (famine protection – seasonal).
- Based on rainfall dependence: supplemental irrigation, full irrigation, pre-sowing (paleo).
- Lift vs flow irrigation.

3. Methods of Application of Irrigation Water

> Water is applied to fields by (A) surface irrigation, (B) sub-surface irrigation, (C) sprinkler irrigation, (D) drip (trickle) irrigation.

❖A. Surface irrigation (flooding)

- Free flooding (wild flooding): water let on field freely, uncontrolled; for cheap, undulating land; wasteful.
- Border strip method: field divided into long strips (border strips) 3–15 m wide, 100–400 m long, separated by ridges; water enters at upper end, flows as a sheet; for wheat, fodder, close-growing crops; slope 0.2–1%.
- Check basin method: field divided into small level basins with bunds; water ponded to required depth; best for flat, permeable soils; paddy; orchards; high efficiency.
- Furrow irrigation: small channels (furrows) between crop rows (30–90 cm spacing); water flows along furrows and seeps laterally into ridges; for row crops – maize, sugarcane, cotton, potato, vegetables; less evaporation, wetting limited. Types: straight, contour, corrugation.

- Basin irrigation (ring basin) for orchards (individual trees).

❖B. Sub-surface irrigation

- Water applied below the ground to raise the water table (natural sub-irrigation) or through buried perforated pipes/ tile drains – used in sandy soils with high water table; low evaporation.

❖C. Sprinkler irrigation

- Water pumped under pressure (2–4 kg/cm^2) through pipes & rotating nozzles, sprayed like artificial rain.
- Suitable for: undulating land, sandy/ highly permeable soils, shallow soils, water-scarce areas, frost protection.
- Advantages: saves 30–50% water, no land levelling, uniform, fertigation, efficiency 70–80%. Disadvantages: high initial cost (pumps, pipes), wind affects uniformity, energy cost, not suited to heavy clay, evaporation in hot wind.

❖D. Drip (trickle) irrigation

- Water delivered drop by drop directly to the root zone through emitters (drippers) on laterals fed by mainline, sub-main, filters, fertiliser tank, pump; discharge 2–8 L/h.
- Efficiency 90–95%; saves 40–70% water; reduces weeds, fertigation, can use saline water, works on any terrain, increases yield; best for orchards, vegetables, sugarcane, cotton. Disadvantages: high cost, clogging of emitters (filtration), rodent/ UV damage, salt build-up at edges.
- Micro-irrigation = drip + micro-sprinklers (PMKSY – "per drop more crop").

❖Comparison

Method	Efficiency	Best for
Border/ check	60–70%	Wheat, paddy, flat land
Furrow	60–75%	Row crops
Sprinkler	70–85%	Undulating land, sandy soil
Drip	90–95%	Orchards, vegetables, water-scarce

4. Quality of Irrigation Water

- Total dissolved salts (TDS)/ salinity: measured by electrical conductivity EC (dS/m or mmhos/cm).

Class	EC (µS/cm)	Suitability
C1 – Low	< 250	Excellent
C2 – Medium	250–750	Good
C3 – High	750–2250	Fair, salt-tolerant crops
C4 – Very high	> 2250	Poor

- **Sodium hazard – Sodium Adsorption Ratio (SAR):**

$$SAR = Na^{+} / \sqrt{[\,(Ca^{2+} + Mg^{2+})/2\,]}\ \text{(all in milli-equivalents/ litre)}$$

- SAR < 10 low hazard; 10–18 medium; 18–26 high; > 26 very high (soil dispersion, permeability ↓).
- **Percent sodium** (SSP) = (Na + K)×100/(Ca + Mg + Na + K); **> 60% harmful.**
- **Residual sodium carbonate** (RSC) = $(CO_3 + HCO_3) - (Ca + Mg)$: < 1.25 safe; > 2.5 unsuitable.
- **Toxic ions:** boron (> 2 mg/L toxic), chloride, fluoride; **heavy metals** in sewage water; pH 6.5–8.4 normal.
- **Suspended silt** (fertile but clogs), **pathogens/ organic load** (sewage – treated only), temperature.
- **Salt tolerance:** barley, cotton, sugar beet are tolerant; wheat, maize, rice are medium (semi-tolerant); beans, citrus and other fruit trees are sensitive.

5. Duty and Delta

Duty (D) = area of land (in hectares) that can be irrigated by a unit discharge of water (1 cumec = 1 m^3/s) running continuously throughout the **base period B** of the crop. Units: **ha/ cumec.**

Base period (B) = time (days) from first watering (preparatory) to last watering before harvest. **Delta (Δ)** = **total depth of water required by a crop during its base period** (depth if spread over the field) – in cm or m.

❖Relation between duty and delta

- 1 cumec for B days gives volume = 1 × B × 86400 m^3. This irrigates D hectares = D × 10^4 m^2:

$$\Delta = 8.64\,B / D\ (\Delta \text{ in metres; } B \text{ in days; } D \text{ in ha/cumec})\ [\Delta(cm) = 864\,B/D]$$

- **Derivation:** $\Delta = (B \times 86400\ m^3) / (D \times 10^4\ m^2) = 8.64\,B/D$ metres.

- Duty $\propto 1/\Delta$: higher duty means less water needed per hectare (better).

❖Duty at various places (not same everywhere because of losses)

- Duty at the field (at the field/ plot): discharge measured at the field – gives the highest duty value.
- Duty at the outlet / head of water course: discharge at the chak outlet – intermediate.
- Duty at the head of the canal (gross duty): discharge at canal head includes seepage & evaporation losses in channels, so for the same area the discharge is largest → duty is smallest.
- Hence $D_{field} > D_{outlet} > D_{head}$ (e.g., rice 800 ha/cumec at the field vs 600 at the canal head).

❖Example

> Wheat: $B = 120$ days, $\Delta = 40$ cm $= 0.4$ m → $D = 8.64 \times 120/0.4 = 2592$ ha/cumec. Rice: $B = 140$ days, $D = 900$ ha/cumec → $\Delta = 8.64 \times 140/900 = 1.34$ m.

> Canal discharge at outlet = 2 cumec, duty at outlet 2000 ha/cumec → area = 4000 ha.

❖Typical duties (ha/cumec at head)

Crop	Base period (d)	Duty
Rice	120–140	800–1000
Wheat	120	1800–2500
Sugar cane	320	700–900
Cotton	180	1400
Maize/ vegetables	100	1200

6. Factors Affecting Duty & Methods of Improving Duty

❖Factors affecting duty

- Type of crop & its base period; climate (temperature, humidity, rain, wind); soil type & permeability (sandy → low duty); method of irrigation (flood vs drip); canal conditions – unlined (seepage losses), length of canal, evaporation; characteristics of the canal system; skill of cultivators & water management; time & frequency of application; tillage/ ploughing; quality of water; topography & slope; water table; rotation of crops; delivery system (rotation, warabandi).

❖Methods of improving duty

- Lining the canals (reduces seepage, 30–40% loss saved); proper canal alignment & maintenance (de-silting, weed control); shorter canal length, better layout.
- Proper land levelling/ shaping, scientific irrigation methods (sprinkler/ drip); crop planning & rotation; conjunctive use of surface & ground water.
- Rotational water supply (warabandi), volumetric water charges, farmers' participation (Water Users' Associations – WUA), canal automation; deep ploughing & mulching; use of high-yielding, short-duration varieties; cut-off of canals during rain; ensure timely supply.

7. Water Requirement of Crops – Types of Soils

> Soil = the loose upper layer of earth crust formed by weathering of rocks + organic matter, supporting plant growth (consists of mineral particles, organic matter, water, air).

❖Types of soil (by texture – particle size)

- Gravel (> 2 mm), Sand (2 – 0.06 mm: coarse/ fine, high permeability, low water retention), Silt (0.06 – 0.002 mm), Clay (< 0.002 mm: high retention, low permeability, swells), Loam (mixture – ideal for agriculture: ~40% sand, 40% silt, 20% clay).
- Textural triangle classifies sandy loam, silty loam, clay loam etc.

❖Indian agricultural soils

Soil	Region	Features	Crops
Alluvial	Indo-Gangetic plains, deltas (Krishna, Godavari)	Fertile, loamy, rich in potash	Rice, wheat, sugarcane
Black (Regur/ cotton)	Deccan (Maharashtra, Telangana, A.P., Gujarat)	Clayey, retains moisture, swells/ cracks, rich in lime/ Fe, poor N,P	Cotton, sorghum, soybean
Red	Rayalaseema, Tamil Nadu, Odisha, Karnataka	Light, porous, acidic, poor N, P, lime	Millets, groundnut, pulses
Laterite	Western Ghats, Kerala, Assam	Leached, acidic, low fertility	Tea, coffee, cashew, rubber
Desert (arid)	Rajasthan	Sandy, saline	Bajra, millets (with irrigation)
Mountain/ forest	Himalayas	Humus-rich, acidic	Tea, fruits
Saline/ alkaline (Usar, Reh)	Punjab, UP, Haryana, canal commands	High salts, poor drainage	After reclamation

- A.P./ Telangana: red soils (≈ 50%), black soils (≈ 25%), alluvial deltas, sandy coastal soils.

8. Preparation of Land for Irrigation

- Clearing of trees, bushes, stones; land levelling & grading (laser levelling) to uniform slope for even flow; ploughing/ tilling to loosen soil, improve infiltration; land shaping – layout of channels (field channels, ditches), border strips, check basins, furrows; bunding (field boundaries); drainage provision; sowing with seed-bed preparation; fertiliser & manure application.
- Objectives: uniform distribution, minimum losses, high efficiency, prevent erosion & water-logging.

9. Soil Fertility & Soil–Water–Plant Relationship

❖Soil fertility

> Fertility = ability of soil to supply nutrients in adequate amount & balance for crop growth. Macro-nutrients: N, P, K (primary), Ca, Mg, S; micro-nutrients: Fe, Zn, Mn, Cu, B, Mo, Cl.

- Maintained by **manures, fertilisers, crop rotation (legumes fix N), green manure, proper irrigation; soil testing, pH 6.5–7.5 ideal; fertigation.**

❖Soil–water–plant relationship

- **Soil-water phases:** solids (45%), water (25%), air (25%), organic (5%) – ideal loam.
- Water in soil is held against gravity by **adhesion & cohesion** (surface tension) in **films & capillary pores;** plant roots absorb **capillary water (available water).**
- **Types of soil water:**
  - **Gravitational water** – drains freely by gravity (after irrigation/ rain); not available (lost in ~ 2–3 days).
  - **Capillary water** – held in pores by surface tension; **available to plants.**
  - **Hygroscopic water** – thin film on soil particles by adhesion; **unavailable.**
  - **Chemically combined water** (in minerals).
- **Moisture content** (dry-weight basis) $w = (W_w / W_s) \times 100$; volume basis $\theta = V_w/V$.

10. Vertical Distribution of Soil Moisture

- After irrigation/ rain: **zones from top: soil–water (root) zone → intermediate zone → capillary fringe → saturated zone.**
- Moisture is **highest at the surface just after wetting,** decreases with depth; **plants extract mostly from the upper root zone (≈ 40% from 1st quarter, 30%, 20%, 10% from the lower quarters of root depth – "40-30-20-10 rule").**
- Root depth: shallow (rice, vegetables 30–60 cm), medium (wheat, maize 90–120 cm), deep (cotton, sugarcane, 1.5–2 m).

11. Soil Moisture Tension & Stress

- **Soil moisture tension (suction) = force (negative pressure) with which water is held by the soil;** expressed as **atmosphere, bar, cm of water, kPa, or pF** ($pF = \log_{10}$ of tension in cm of water).

> Tension ↑ as soil dries; saturation ≈ 0 ; field capacity ≈ 1/3 atm (pF 2.5) ; PWP ≈ 15 atm (pF 4.2)

- Measured by **tensiometer** (porous cup, vacuum gauge, range 0–0.8 atm), gypsum blocks, neutron probe, pressure-plate apparatus.

- Soil moisture stress = combined effect of tension & osmotic (salt) forces that reduces availability of water to plants; plant wilts when stress exceeds root suction.
- Total stress (potential) = matric (tension) + osmotic + gravitational components.

12. Soil Moisture Constants

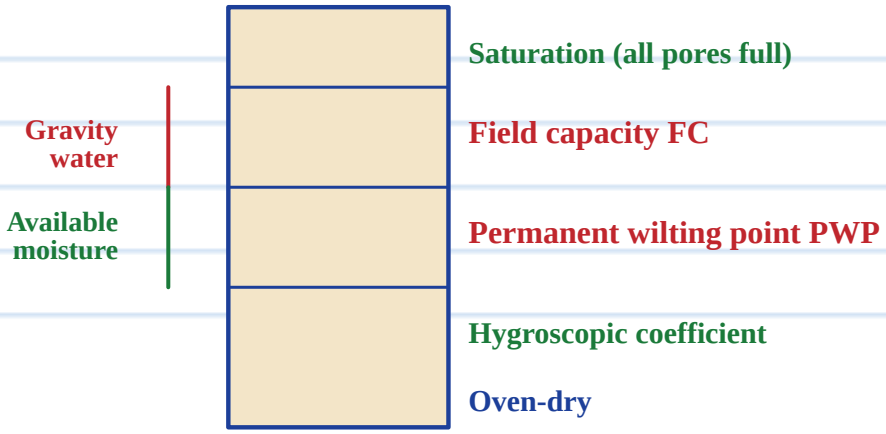


Fig 3.1 Soil moisture constants

- Saturation capacity: all pores filled; maximum water content (moisture tension ≈ 0).
- Field capacity (FC): water held by soil after free (gravitational) drainage has ceased (≈ 2–3 days after irrigation); tension ≈ 1/3 atm; upper limit of available water.
- Permanent wilting point (PWP / wilting coefficient): moisture content at which plants wilt permanently & cannot recover (tension 15 atm); lower limit of available water.
- Available moisture (AM) = FC – PWP (readily available portion ≈ 50–75% of it).
- Hygroscopic coefficient: moisture absorbed from saturated air, held at tension ≈ 31 atm; ultimate wilting point ≈ hygroscopic.
- Moisture equivalent, optimum moisture, readily available moisture.

❖Typical values (% of dry weight)

Soil	FC	PWP	Available
Sand	10	4	6
Sandy loam	18	8	10
Loam	27	13	14
Clay loam	35	18	17
Clay	40	24	16

❖Limiting soil-moisture conditions

- Upper limit = field capacity (beyond this water is wasted by percolation).
- Lower limit = permanent wilting point; in practice irrigation is given when 50–60% of available moisture is used – "optimum moisture/ allowable depletion (p)" (before reaching PWP; crop yield falls beyond "critical moisture").
- Readily available moisture = 0.5–0.75 × (FC – PWP).
- Over-irrigation → aeration problem (waterlogging); under-irrigation → stress.

13. Depth & Frequency of Irrigation

Depth of water per irrigation $d = [(F_c - PWP)/100] \times (\gamma_d/\gamma_w) \times D \times p$
(or $d = (F_c - M_0)/100 \times \gamma_d/\gamma_w \times D$)

- F_c = field capacity (%), PWP (%), γ_d = dry unit weight (specific gravity) of soil, γ_w = unit weight of water, D = root-zone depth, p = allowable depletion fraction (0.5–0.6), M_0 = moisture at which irrigation is given.

Frequency (interval) of irrigation $f = d / C_u$ (days), C_u = peak consumptive use (mm/day)

❖Example

Root zone D = 80 cm, F_c = 28%, PWP = 14%, bulk density ratio γ_d/γ_w = 1.4, p = 0.5 → $d = 0.14 \times 1.4 \times 80 \times 0.5 = 7.84$ cm. Peak use = 0.5 cm/day → f = 7.84/0.5 ≈ 15 days.

14. Command Area Terms

- Gross Command Area (GCA): total area enclosed by the boundaries that can be irrigated by a canal system, whether or not cultivable (includes roads, villages, ponds, barren land).
- Culturable Command Area (CCA): the part of GCA that is fit for cultivation (excludes unfit land, habitation). CCA = GCA – unculturable area.
- Culturable cultivated area: CCA actually cultivated at a time (sown). Culturable uncultivated area: CCA not cultivated (fallow/ waste, but fit). Intensity of irrigation = irrigated area/ CCA × 100%.

GCA = CCA + Unculturable area ; CCA = Cultivated + Uncultivated (culturable)

- Irrigation capacity of canal designed on CCA × intensity; outlet serves a chak.

15. Kor Depth and Kor Period

> Kor watering = the first watering (usually applied to a crop when it is a few cm tall) – generally the heaviest and given to soften soil & for germination.

- Kor depth = depth of water applied for the first watering (e.g., wheat ≈ 13.5 cm, rice ≈ 19 cm, sugar cane ≈ 16.5 cm, cotton ≈ 10 cm).
- Kor period = time (days) for which the first (kor) watering is applied/ the period during which kor watering is given in the whole canal command (wheat ≈ 4 weeks; rice ≈ 2.5 weeks) – the canal is designed on the peak kor requirement.

Outlet discharge $Q = A \times d / (B_{kor}) \approx (A \times \text{Kor depth}) / (\text{kor period} \times 86400)$

❖Example

> Wheat area A = 2000 ha, kor depth = 13.5 cm, kor period 4 weeks (28 days): Volume = $2000 \times 10^4 \times 0.135 = 2.7 \times 10^6\ m^3$; $Q = 2.7 \times 10^6 / (28 \times 86400) = 1.12\ m^3/s$.

16. Crop Seasons & Crop Rotation

❖Crop seasons in India

Season	Sowing – harvest	Crops
Kharif (monsoon)	June–July → Sept–Oct	Rice (paddy), maize, jowar, bajra, cotton, groundnut, sugarcane (annual)
Rabi (winter)	Oct–Nov → Mar–Apr	Wheat, barley, gram, peas, mustard, sunflower
Zaid (summer)	Mar → June	Watermelon, cucumber, vegetables, fodder

- Perennial/ two-seasonal crops: sugarcane (12–18 months), banana, cotton (8 months).
- Different crops have different water requirement (paddy 1200–2500 mm; wheat 450–650; sugarcane 1500–2500; cotton 700–1300; maize 500–800; pulses 250–500 mm).

❖Crop rotation

> Crop rotation = growing different crops in succession on the same land in a planned sequence.

- **Advantages:** maintains soil fertility (legumes fix nitrogen), controls pests, weeds and diseases, balances nutrient removal & root depth (shallow/ deep rooted), improves soil structure, reduces salinity, reduces water requirement variation, better yields, risk spreading.
- **Examples:** rice–wheat, rice–pulses, cotton–groundnut, paddy–groundnut–green gram, wheat–fallow–sugarcane. Fallowing, mixed & inter-cropping.

17. Irrigation Efficiencies

> Efficiency = ratio of output water to input water at various stages (water is lost by seepage, evaporation, deep percolation, runoff).

Conveyance (canal) efficiency $\eta_c = W_f / W_r \times 100$ (water delivered to the field / water diverted at river head)

Water application efficiency $\eta_a = W_s / W_f \times 100$ (water stored in root zone / water delivered to the field)

Water storage efficiency $\eta_s = W_s / W_n \times 100$ (stored / water needed to bring root zone to FC)

Water use efficiency $\eta_u = W_u / W_d \times 100$ (consumed by crop (consumptive use)/ water delivered)

- Project (overall) efficiency = $\eta_c \times \eta_a$ (≈ 40–60% for flood systems; drip 90%).
- Water distribution efficiency $\eta_d = [1 - (y/d)] \times 100$ (y = average numerical deviation from mean depth d).
- Typical: η_c 60–80%; η_a 50–70% (surface), 75% sprinkler, 90% drip.

❖Example

> Water diverted at head = 5 m^3/s, delivered to field = 3.5 m^3/s → η_c = 70%.
> Water stored in root zone = 2.8 → η_a = 2.8/3.5 = 80%; overall = 0.7×0.8 = 56%.

18. Determination of Irrigation Requirements of Crops

❖Terms

- **Consumptive use (C_u)/ Evapotranspiration:** water used by crop in transpiration + building plant tissue + evaporation from soil/ leaf surface.

- Effective rainfall (R_e): portion of rain stored in root zone & useful (≈ 70–80% of rain above 5–6 mm; 0 if too little/ too much).
- Net irrigation requirement (NIR) = C_u – R_e – (soil moisture & ground-water contribution) + leaching requirement (if saline).

> NIR = C_u – R_e – S; Field irrigation requirement FIR = NIR / η_a; Gross irrigation requirement GIR = NIR / (η_a η_c)

- Add water for land preparation (puddling – paddy: 150–250 mm), nursery, percolation losses (paddy: 2–6 mm/day), leaching.

❖Assessment of irrigation water

- Water is charged to cultivators by:
  - Volumetric (by volume) assessment – metering at outlet (fairest; modules like Gibb's, Kennedy's).
  - Area (acreage) assessment – crop-wise rate per hectare (most common in India; rates depend on crop & season, e.g., paddy higher than dry crop).
  - Seasonal/ per-irrigation rates, permanent assessment (fixed per ha irrespective of use, whether or not used), Batai (share of crop), block system, Warabandi (fixed rotation turn schedule by time).

❖Warabandi (rotational water supply) & water charges

- Fixed day/ hour schedule for each farmer – equitable distribution, reduces conflict & wastage; used in Punjab/ Haryana; A.P. "Neeru-Pragathi" water management under WUAs.

19. Consumptive Use of Water

> Consumptive use (C_u) = total water (depth) needed by a crop from sowing to harvest for transpiration + evaporation (evapotranspiration, ET). Expressed as depth (mm/ day, mm/ season) and used to find delta.

❖Factors affecting consumptive use

- Climate: temperature, humidity, wind, sunshine hours/ solar radiation, rainfall.
- Crop: type, growth stage (peak at flowering), crop period, leaf area, density, root depth.
- Soil: moisture availability, soil texture & water table depth, fertility, salinity.
- Cultural/ management: irrigation method & frequency, tillage, mulch, planting date.

- Latitude, season, altitude.

❖Direct measurement

- Tank (lysimeter) experiments: crop grown in large tank/ drum with known water inputs; ET = water added – drained (lysimeter weighing).
- Field plot experiments: measure water applied + rainfall + change in soil moisture – drainage.
- Soil-moisture studies (gravimetric/ neutron probe) in the root zone; integrating method (water balance in command area).
- Inflow–outflow studies for large areas.

❖Determination by equations (theory)

- (1) Blaney–Criddle formula (USBR):

$$C_u = \Sigma\, k\, f,\ f = p\,(0.46\,T + 8.13)\ \text{(mm/day for daily p)}$$

  - T = mean temperature (°C); p = % of annual daytime hours occurring in the period (≈ 0.27 per day, 8–9 per month); f = consumptive-use factor; k = crop coefficient (e.g., 0.8–1.2 for different crops/ season).
- (2) Penman equation (energy balance + aerodynamic) – $ET_0 = [\Delta\,(R_n - G) + \gamma \cdot 6.43\,(1 + 0.536\,u)\,(e_s - e_a)\,] / [\lambda\,(\Delta + \gamma)\,]$; FAO Penman–Monteith (standard): ET_0 reference crop; $ET_c = K_c \times ET_0$.
- (3) Hargreaves: $ET_0 = 0.0023\,R_a\,(T + 17.8)\,(T_{max} - T_{min})^{0.5}$ (needs only temperature).
- (4) Thornthwaite, Christiansen, pan-evaporation method: $ET_0 = K_p \times E_{pan}$ ($K_p \approx 0.7$–0.8).
- (5) Lowry–Johnson, Hargreaves–Samani, Jensen–Haise.

❖Example

Blaney–Criddle (daily form, p = daily % of annual daytime hours ≈ 0.27): T = 28 °C → f = 0.27 (0.46×28 + 8.13) = 0.27 × 21.0 = 5.67 mm/day; k = 0.9 → C_u = k f = 5.1 mm/day (≈ 612 mm for a 120-day crop).

Simple calculation: crop period 120 days, ET_0 average 5 mm/d, K_c = 1.0 → C_u = 600 mm; effective rain 150 mm → NIR = 450 mm; with η_a = 60% & η_c = 75% → GIR = 450/(0.6×0.75) = 1000 mm.

20. Solved Problems

❖Problem 1 – Delta

> $B = 110$ days, $D = 1600$ ha/cumec → $\Delta = 8.64 \times 110/1600 = 0.594$ m ≈ 59.4 cm.

❖Problem 2 – Canal discharge for CCA

> CCA = 10,000 ha under wheat (delta 45 cm, base 120 d): $D = 8.64 \times 120/0.45 = 2304$ ha/cumec → $Q = 10000/2304 = 4.34$ m^3/s (at outlet).

❖Problem 3 – Efficiency-based requirement

> Net field requirement = 6 cm over 100 ha = 6000 m^3; $\eta_a = 60\%$ → delivered = 10,000 m^3; $\eta_c = 75\%$ → at head = 13,333 m^3.

❖Problem 4 – Frequency

> $d = 6$ cm, peak $C_u = 0.6$ cm/day → $f = 10$ days.

21. Important Questions – Unit III

- Necessity, importance, advantages & ill effects of irrigation. Types of irrigation.
- Describe methods of application: border, check basin, furrow, sprinkler, drip.
- Quality of irrigation water (SAR, EC); define duty & delta and derive $\Delta = 8.64B/D$; factors affecting duty & methods of improving.
- Soil moisture constants; depth & frequency of irrigation; GCA, CCA, kor depth/ period.
- Crop seasons & rotation; irrigation efficiencies; consumptive use & equations; assessment of irrigation water.

$$\Delta = 8.64B/D \; ; \; d = (Fc - PWP)(\gamma d/\gamma w)D \; ; \; f = d/Cu \; ; \; GIR = NIR/(\eta a \, \eta c)$$

WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – IV : CHANNELS – SILT THEORIES, WATER LOGGING & CANAL LINING

1. Classification of Canals

> A canal is an artificial channel constructed to carry water from a source (river/ reservoir) to the command area.

❖(a) Based on nature of source

- Permanent (perennial) canals: draw from perennial rivers with head works (weir/ barrage) or reservoirs – water throughout the year (Godavari, Krishna delta canals, Nagarjuna Sagar canals).
- Inundation canals: taken off without head works, flow only when river is in flood (high stage); mostly in Punjab/ Sind historically.

❖(b) Based on function

- Irrigation, carrier, feeder, navigation, power (hydro) and drainage canals.

❖(c) Based on discharge & position in the network

- Main canal (carries entire supply, no direct irrigation – no outlets), branch canals (> 5 cumec), major distributaries (0.25–5 cumec), minor distributaries (< 0.25), water courses (field channels – owned by cultivators).

❖(d) Based on alignment

- Ridge (watershed) canal, contour canal, side-slope canal.

❖(e) Based on lining & construction

- Lined vs unlined; in cutting, in filling, partial cutting & filling.

❖(f) Based on financial output

- Productive (revenue > cost) and protective (famine relief).

2. Canal Alignment

> Alignment = the route/ layout of the canal (plan) selected for economical, efficient command.

❖Alignment types

- Ridge canal (watershed canal): along the ridge line between two drainage basins → commands area on both sides, no cross-drainage works, minimal cutting & filling – best, most economical.
- Contour canal: follows a contour with a small slope, commands area on one side only (down slope); cross-drainage works needed; used in hilly regions.
- Side-slope canal: runs perpendicular to contours (down the slope); no cross-drainage problems, but silting/ scour issues; rare.

❖Requirements for good alignment

- Commanding position (FSL above ground level of the area → gravity flow); shortest length; minimum cross-drainage works; balanced cutting & filling (balancing depth); avoid rocky strata, sandy/ porous soil, villages, valuable land, historic places; minimum curves (radius ≥ 5–6 × bed width); proper drainage & water-logging avoidance; head regulator at suitable place; easy access for maintenance; no unnecessary diversions.

3. Inundation Canals

- No permanent diversion works; water enters when river water level exceeds canal bed; flows during monsoon floods only → cultivation of rabi crops with flood-moisture; silt-laden fertile water deposits.
- Advantages: cheap (no weir), natural silt manuring. Disadvantages: uncertain & seasonal, heavy silting, damage by flood, irregular supply, not for perennial crops.

4. Cross-section of an Irrigation Channel

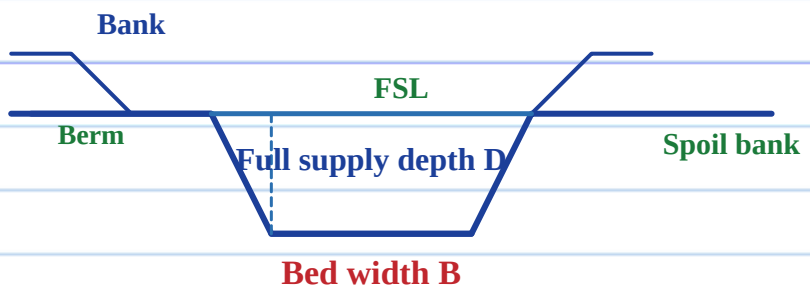


Fig 4.1 Cross-section of an irrigation channel (water level = FSL)

❖Terms

- Full supply level (FSL): water surface in design discharge; full supply depth D; bed width B; bed level; side slopes (cutting 1:1 – 1.5:1 H:V; filling 1.5:1 – 2:1; in sandy 2:1).
- Berm: horizontal strip of land (≥ 1–4 m) left between toe of bank and edge of cut (or borrow pit) for stability, collect scouring, protect bank, space for silt, future widening.
- Freeboard: vertical distance bank top – FSL (0.5 – 1.0 m for small, up to 1.5 m for large canals) – to prevent overtopping.
- Banks: embankments on both sides; top width (service road 5–7.5 m on one side, 3–4 m on the other); inspection road.
- Dowel (Dowla): small bank on road side to protect the canal from rain-cut, 0.3–0.5 m high, 1–2 m wide; backing/ berm.
- Hydraulic gradient line (HGL) & saturation line (the line of seepage through bank; keep it within the bank section at safe slope 4:1 to 6:1).

❖Types of canal section (by relation to ground)

- Fully in cutting (deep, costly; avoids bank), fully in filling (on embankment – seepage danger), partially cutting & partially filling (balanced) – most economical.

❖Balancing depth

> Balancing depth = the depth of cutting at which the excavated earth just equals the earth required for forming the banks (embankment) – i.e. quantity of cutting = quantity of filling; gives minimum cost (no spoil bank or borrow pit).

- Depends on bed width, side slopes, bank top width, freeboard, FSL above ground. Found by trial/ using table or formula (e.g., for trapezoidal section: cut area = fill area).

❖Borrow pit

> Borrow pit = excavation (pit) made on the ground outside the canal to obtain earth for constructing the banks when excavation from the canal is insufficient (canal in filling). Should be dug on the canal-side (inside) of the land, not close to the bank toe (leave berm ≥ 3–5 m), kept shallow and drained.

❖Spoil bank

> Spoil bank = embankment formed from surplus excavated earth (spoil) when the canal is in deep cutting, deposited as mounds on one/ both sides near the canal (30 – 60 cm wide; berm ≥ 3 m). Use it as protective bund, plantation; placed only when excavation > filling. Must have gaps/ cross-sections for drainage & avoid blocking surface drainage.

❖Land width

> Land width = total width of land acquired for the canal (including bed, banks, berms, borrow pits, spoil banks, service roads, drains). Land width = bed width + 2 × (side slopes × depth) + 2 berms + 2 × bank top widths + extra for borrow pits/ spoil banks. Governed by the irrigation department's land-acquisition schedule.

5. Silt Theories – Introduction

- Canals carry silt (sediment) from the river; if velocity is too low → silting (deposition) reduces capacity; too high → scouring (erosion of bed & banks). Aim: design a non-silting, non-scouring (regime) channel.
- Regime channel: a channel that neither silts nor scours – stable cross-section & slope over years.
- Types of regime: initial regime (bed silt balanced – earth channels that attained stability), final (true) regime (all variables – discharge, silt charge, silt grade – constant; the channel is actually regime; practically only approached).
- Theories: Kennedy (1895, Upper Bari Doab canal); Lacey (1930s – regime theory); Gibbs, Blench, Garde etc.
- Silt supporting power – via eddies generated at the bed (vertical component of turbulence) keep silt in suspension.

6. Kennedy's Silt Theory

> Kennedy: "The silt in a channel is kept in suspension by the vertical components of the eddies generated by the friction of flowing water at the bed." The critical velocity V_0 is the mean velocity that makes the channel neither silt nor scour.

$V_0 = 0.55\ m\ D^{0.64}$ (V_0 in m/s; D = depth of water in channel, m)

- m = critical velocity ratio (C.V.R.) = V/V_0 – depends on silt grade: m = 1.0 for ordinary silt, 0.7–0.8 for very fine silt, 1.1–1.2 for coarse sandy silt. If m = 1 → no silting/ scour; m > 1 → scour; m < 1 → silt.
- Kutter's formula for velocity: $V = [(23 + 0.00155/S + 1/n) / (1 + (23 + 0.00155/S)\ n/\sqrt{R})]\sqrt{(RS)}$.
- Or Manning's: $V = (1/n)\ R^{2/3}\ S^{1/2}$ (modern).
- Also Kennedy's B/D ratio (bed width/ depth) taken from tables / by Garret's diagrams (graphical) to avoid trial (Garret).

7. Kennedy's Method of Channel Design

❖Given: discharge Q, silt factor m, rugosity n (0.0225), slope S (or B/D ratio).

❖Procedure – when S is known

- 1. Assume depth D (trial). 2. $V_0 = 0.55\ m\ D^{0.64}$. 3. Area $A = Q/V_0$. 4. Choose side slope (½:1 or 1:1 H:V); bed width B from $A = (B + D)\ D$. 5. Find $R = A/P$, wetted perimeter $P = B + 2.83\ D$ (side slope 1:1) or $B + 2.236\ D$ (½:1). 6. Compute $V = (1/n)\ R^{2/3}\ S^{1/2}$ (or Kutter). 7. If $V = V_0$ (within small difference) → design OK; else revise D and repeat.

❖Procedure – when B/D ratio is given

- 1. Assume D; $B = (B/D) \times D$, $A = (B + D)D$ (rect.) or $(B + zD)D$. 2. $V = Q/A$; 3. $V_0 = 0.55\ m\ D^{0.64}$; 4. If $V \neq V_0$, adjust D & repeat; 5. Then find slope S from Kutter/ Manning: $S = (V\ n / R^{2/3})^2$.

❖Example

> Design a channel for $Q = 10\ m^3/s$, m = 1, n = 0.0225, S = 1/5000, side slopes 1:1. Try D = 1.5 m: $V_0 = 0.55 \times 1.5^{0.64} = 0.55 \times 1.296 = 0.713$ m/s; $A = 10/0.713 = 14.03\ m^2$; $B = A/D - D = 9.35 - 1.5 = 7.85$ m; $P = B + 2.83\ D = 7.85 + 4.24 = 12.09$ m; $R = 14.03/12.09 = 1.16$ m; $V = (1/0.0225)\ 1.16^{2/3}\ (0.0002)^{1/2} = 44.4 \times 1.104 \times 0.01414 = 0.69\ m/s \approx V_0$ ✔ (accept; revise D to 1.47 for exact).

8. Drawbacks of Kennedy's Theory

- Silt grade treated only by empirical factor m; no definite relation with silt size (silt not defined); m is arbitrary.
- No equation for B/D or slope → trial & error (needs Garret's/ Lacey's supplement); B/D ratio arbitrary.
- Kutter's n chosen arbitrarily; no allowance for silt charge (quantity) or bed load.
- Based on only 22 channels of Upper Bari Doab canal (Punjab, a particular type of silt) – not general.
- Assumes silt is held by vertical eddies only – ignores concentration & water-silt interaction; considers bed width and depth independent; side slopes not studied.
- Ignores effect of sides (banks) on eddies – formula only for bed.
- Fails for very different silt, large canals.

9. Lacey's Regime Theory

Lacey: channel is in "regime" when it flows in incoherent alluvium of the same grade as that carried, with constant discharge & silt charge. Silt is held in suspension by the vertical components of eddies generated at ALL points of the wetted perimeter (bed & sides); there are three regimes – initial, final (true), and "in regime" conditions (final regime = cross-section & slope become constant, with the channel having a flow condition of semi-elliptical shape).

- Silt factor $f = 1.76\sqrt{d_{mm}}$ (d = mean particle size in mm; f = 1 for 0.55 mm; $f \approx 0.7–1.5$ for typical silt).

❖Lacey's regime equations (SI)

$$V = (Q f^2 / 140)^{1/6} \text{ (m/s)}$$

$$A = Q / V;\ P = 4.75\sqrt{Q} \text{ (wetted perimeter, m)}$$

$$R = 5 V^2 / (2 f) \text{ (hydraulic mean radius, m)}$$

$$S = f^{5/3} / (3340\, Q^{1/6}) \text{ (bed slope)}$$

- Other relations: $f = 1.76\sqrt{d}$, $V = 10.8\,R^{2/3}\,S^{1/3}$, $Q = A\,V$, $R = A/P$.
- Lacey's scour depth: $R_{scour} = 0.47\,(Q/f)^{1/3}$.

❖ **Lacey's design procedure (given Q and d (or f))**

- 1. Find silt factor $f = 1.76\sqrt{d_{mm}}$ (or given).
- 2. $V = (Q\,f^2/140)^{1/6}$.
- 3. $A = Q/V$.
- 4. $R = 5V^2/(2f)$.
- 5. $P = 4.75\sqrt{Q}$ (or $P = A/R$).
- 6. Using $P = B + 2.236\,D$ and $A = (B + 0.5\,D)\,D$ (side slope ½:1), solve for depth D and bed width B (quadratic).
- 7. $S = f^{5/3}/(3340\,Q^{1/6})$.

❖ **Example**

> Design by Lacey: $Q = 30\ m^3/s$, $d = 0.5$ mm. $f = 1.76\sqrt{0.5} = 1.245$. $V = (30\times1.245^2/140)^{1/6} = (0.3321)^{1/6} = 0.832$ m/s; $A = 30/0.832 = 36.06\ m^2$; $R = 5\times0.692/(2\times1.245) = 1.39$ m; $P = 4.75\sqrt{30} = 26.0$ m (also $P = A/R = 25.9$ ✔); $S = 1.245^{5/3}/(3340\times30^{1/6}) = 1.44/(3340\times1.763) = 1/4090 \approx 0.000245$.

10. Defects in Lacey's Theory

- Equations are empirical, derived from Indian & Egyptian canals (not universal); no rational base.
- Silt factor f is not a true property of silt; relation with d is empirical; concept of "regime" is ill-defined; true regime hardly ever reached.
- Silt charge (concentration) is ignored; only depends on Q & f; B/D is not independent (side slopes not accounted).
- Lacey's "eddy" concept (generated at all points) is unproven; hydraulic mean radius R relation doubtful; the sides of the channel treated as same as bed.
- $P = 4.75\sqrt{Q}$ relation shows P depends only on Q – regardless of bank soil & silt → inconsistent for stronger banks (rock/ cohesive).
- Not applicable to lined channels, rivers with bed load, very large/ small canals.

11. Comparison of Kennedy's and Lacey's Theories

Point	Kennedy	Lacey
Silt supported by	Vertical eddies from bed only	Eddies from bed and sides (whole perimeter)
Basic criterion	Critical velocity V_0	Regime velocity V
Silt grade	Critical velocity ratio m (arbitrary)	Silt factor $f = 1.76\sqrt{d}$ (from size)
Regime types	One (non-silting/ non-scouring)	Initial, final & true
Equations	One: $V_0 = 0.55\ m\ D^{0.64}$	Three (V, P, S) – no trial & error
Design	Trial & error (Garret's curves)	Direct with equations
Slope	Not given	Given
B/D ratio	Not given (assumed)	Obtained from regime equations
Based on data	Upper Bari Doab canals only	Many canals & rivers (more general)
Channel shape	Trapezoidal (any)	Semi-elliptical (final regime)
Silt charge	Not considered	Not considered

12. WATER LOGGING

> Water-logging = a condition in which the soil pores in the root zone become saturated with water due to rise of the water table (to within 1.5 m of ground surface in the root zone), so air circulation stops and plants suffer. (Capillary rise brings salts to surface.)

- Zones of water table: < 1.5 m – water-logged (danger); 1.5 – 3 m – critical (capillary-fringe effects); > 3 m – safe. Water-logged area in India ≈ 6 M ha (canal commands of Punjab, Haryana, Rajasthan, UP, A.P.).

❖Causes of water-logging

- Seepage from unlined canals & water-courses (main cause) and reservoirs/ tanks.
- Over-irrigation (excess application, flooding of fields), intensive irrigation without drainage.
- Inadequate natural drainage (flat topography, obstruction of natural drainage by canals, roads, railway embankments); blocking of drains.

- Impervious soil/ sub-soil layer (hard pan/ clay) near surface preventing percolation; high rainfall, flood; sea water intrusion/ high tide in coastal belts.
- Insufficient outlets/ lack of drainage design, un-planned canal alignment; poor irrigation practices (crop mixes needing little water); lack of pumping.

❖Effects of water-logging

- Depletion of oxygen in root zone → poor root growth, reduced yield, crop failure; anaerobic conditions → toxic substances; reduced soil temperature & bacterial activity (nitrification), delays tillage/ sowing.
- Salinity/ alkalinity (capillary rise deposits salts: "Reh/ Kallar/ Usar" lands); soil becomes sour (acidic/ alkaline).
- Breeding ground of mosquitoes & malaria, water-borne diseases; damage to buildings (dampness), roads, foundations; reduced stability of structures; loss of agricultural area & land value; ecological & economic loss; migration.
- Growth of weeds (e.g., Typha), waterlogged pasture.

❖Remedial (preventive & curative) measures

- Prevention: Reduce seepage – lining of canals & water courses; control of intensity of irrigation; efficient water management (proper depth & frequency, warabandi); proper crop rotation (deep-rooted & low-delta crops); avoid over-irrigation; optimum use of canal water with conjunctive use of ground water; adequate drainage design when planning projects; regulating supply to canals; maintain natural drainage (provide cross-drainage works for obstruction); limit reservoir seepage.
- Curative: Surface drainage – open drains; sub-surface (tile/ pipe) drainage; tube-well pumping (vertical drainage) to lower water table; improving natural drainage – deepening & straightening streams; tile drains/ mole drains; horizontal drains; re-sectioning/ pumping of ground water; biodrainage (Eucalyptus plantation); bunds & dikes.

❖Drainage systems (short)

- Surface drains (open ditches), sub-surface drains (perforated pipes in gravel envelope, spacing by Hooghoudt's equation), mole drains, vertical drainage via tube wells (skimming wells).
- Drain spacing (Hooghoudt): $L^2 = 8\,K\,d\,h\,/\,q + 4\,K\,h^2/q$.

13. Saline and Alkaline Soils & their Reclamation

❖Problem soils

- Saline soils (white alkali, solonchak): EC > 4 dS/m, ESP < 15%, pH < 8.5 – excess soluble salts (chlorides, sulphates of Na, Ca, Mg); white crust; reduce osmotic availability of water.
- Alkaline (sodic) soils (black alkali, solonetz): ESP (exchangeable sodium percentage) > 15%, pH > 8.5 (up to 10), EC < 4; dispersed clay, poor structure/ permeability, dark crust, sodium carbonate.
- Saline-alkaline soils: EC > 4 & ESP > 15.
- Causes: canal seepage & water-logging, high evaporation, brackish irrigation water, poor drainage, coastal sea-water, dry climate.

❖Reclamation

- Saline soils: leaching – apply excess good water (flooding) with drainage to wash salts below root zone; provide drainage (surface/ sub-surface); lower water table; salt-tolerant crops (barley, cotton, sugar beet, date palm); organic manure, mulching, green manure; scraping surface salts; deep ploughing; proper irrigation (frequent light irrigation).
- Alkali soils: chemical amendments to replace Na^+ with Ca^{2+}: gypsum ($CaSO_4 \cdot 2H_2O$) – most common; sulphur, sulphuric acid, pyrite, lime, calcium chloride, molasses, press mud; then leach with water; grow rice/ barley/ dhaincha (green manure), add organic matter.

$$2\text{Na-soil} + CaSO_4 \rightarrow \text{Ca-soil} + Na_2SO_4 \text{ (leached out)}$$

- Gypsum requirement (GR) = based on ESP reduction & CEC.
- Prevention: lined canals, proper drainage, conjunctive use, water-quality monitoring (SAR/ RSC), avoid over-irrigation.

14. Losses in Canals

> Canal water is lost between head and field by:

- Seepage (percolation) losses – through bed & sides; major (≈ 70–90% of total loss); depends on soil permeability, wetted perimeter, water depth, silt content, ground-water level, age of canal.

- Evaporation losses – from water surface (1–2%, ≈ 10–20 mm/day max in summer).
- Absorption (initial wetting) losses – dry canal bed soaks water at start of flow.
- Other: leakage, transpiration by weeds, operational wastage (spill, tail outfall), outlets leakage, gate & structure leakage.

❖Estimation of seepage

- Typical seepage rates (cumec per million m^2 of wetted area): sand 4–6; sandy loam 2.5–3.5; loam 1.5–2.5; clay 1–1.5; concrete-lined 0.3–0.6.
- Empirical formulas (Moritz, Punjab Irrigation Research Institute, Davis–Wilson) relate seepage to discharge, velocity, wetted perimeter and soil type; loss = rate × wetted area (P × L).
- Conveyance (transmission) loss ≈ 30–50% in unlined canals (A.P. average ≈ 30–40%).
- Control: lining, compaction, silt-laden water (natural clogging), reduce length, proper maintenance, bed-width/ depth ratio optimisation (narrow, deep sections lose less).

15. Lining of Irrigation Channels

> Canal lining = impervious (or semi-pervious) layer of concrete, brick, stone, asphalt, soil-cement, polythene/ geomembrane on bed & sides of canal to reduce seepage and erosion.

❖Necessity (why lining?)

- Reduce seepage loss (save water – 30–70%); control water-logging & salinity; prevent weeds; higher velocity allowed (non-scouring) → smaller cross-section & less land; lower maintenance (silting, weed control); protect banks (stability); safe against breaches; increase command area/ duty; more reliable supply, reduces cost of drainage; prevents rat holes.

❖Advantages

- Water saving (≈ 60–90% reduction in seepage); prevents water logging; smaller section → less land acquisition; less silting, no weed growth; permits high velocity (≈ 2 m/s) → smaller canal; lower maintenance cost; stable slopes (steeper side slopes); better discharge (smooth surface, n = 0.013–0.015 vs 0.025); less evaporation due to narrow deep section; more irrigation area per cumec; longer life & no breaches.

❖Disadvantages

- High initial cost (but pays back); repair difficult (cracks); problems of uplift pressure when canal is empty and ground water is high → failure (needs drainage/ pressure relief); not suitable for expansive soils/ frost without proper joints; hinders ground-water recharge; difficult to shift outlets; requires skilled labour & quality control; damage by animals/ people.

❖Types of lining

- (a) Hard-surface lining
  - Cement concrete lining – plain cement concrete (1:3:6 or 1:2:4), thickness 5 – 15 cm (bed & sides; slip-form paving) with expansion & contraction joints (every 3–6 m; filled with bitumen/ sealants) – most common. Also pre-cast concrete slabs/ tiles, shotcrete (gunite) with wire mesh for repairs, reinforced concrete for large canals.
  - Brick lining – single/ double layer of bricks laid in cement mortar over a bed of lime/ cement mortar; 7.5–15 cm; cheap in areas without stone; less durable.
  - Stone-masonry lining (dressed/ random rubble) – where stone is available; strong.
  - Asphaltic (bituminous) lining – asphalt concrete / bituminous membrane, flexible, tolerates settlement; for flexible soils.
- (b) Earth-type (soil) lining
  - Compacted earth lining (0.3–1.0 m of clayey soil), stabilised soil/ soil–cement (5–15% cement mixed with soil), bentonite/ clay blanket – cheap, but not durable against velocity.
- (c) Buried membrane lining – LDPE/ HDPE polythene film or geo-membrane (0.25 – 1 mm) covered by 15 cm soil/ concrete/ brick (protect from sunlight & puncture); cheap, flexible, quick – very popular for small canals.

- (d) Other: geosynthetic clay liner, ferro-cement, precast U-flumes, sprayed bitumen.

❖ **Choice of lining depends on**

- Cost, soil, canal size, velocity, seepage allowed, availability of material, ground-water level.

16. Design of a Lined Canal

Design is by Manning's formula, adopting hydraulically efficient (most economical) section (trapezoidal) with lined (smooth) surface and permissible velocity (non-silting & non-scouring, max 2.0–2.5 m/s for concrete lining).

❖ **Steps**

- 1. Given Q, bed slope S, Manning's n (0.015 for concrete lining; 0.016–0.02 brick), side slope z : 1 (1.5 : 1 for concrete; 1:1 stable soils), max velocity V_{max}.
- 2. Most economical trapezoidal section (best hydraulic section): half top width = side slope length, $R = D/2$; $B = 2D(\sqrt{(1+z^2)} - z)$.
- 3. Area $A = (B + zD)D$; wetted perimeter $P = B + 2D\sqrt{(1+z^2)}$; $R = A/P = D/2$.
- 4. $Q = AV = A(1/n)R^{2/3}S^{1/2}$ → solve D (trial or direct).
- 5. Check $V < V_{max}$; adopt freeboard (≈ 0.6 m for $Q \leq 10\ m^3/s$; 0.75 m up to 30 m^3/s), lining thickness, joints, berm width; design drainage/ pressure-relief valves under the lining.
- 6. Lining thickness t (cm) depends on size: 5 cm (small), 7.5 cm ($Q \leq 14$ cumec), 10–15 cm (> 28 cumec).

❖ **Example**

Design a most-economical concrete-lined trapezoidal canal: $Q = 15\ m^3/s$, $S = 1/5000$, $n = 0.015$, $z = 1.5$.

- Best section: $B = 2D(\sqrt{(1+2.25)} - 1.5) = 2D(1.803 - 1.5) = 0.606D$; $A = (0.606D + 1.5D)D = 2.106D^2$; $P = 0.606D + 2D \times 1.803 = 4.212D$; $R = D/2$.

- $Q = (1/0.015) \times 2.106D^2 \times (D/2)^{2/3} \times (0.0002)^{1/2} = 66.7 \times 2.106 \times 0.63 \times 0.01414 \times D^{8/3} = 1.251\, D^{8/3} = 15 \rightarrow D^{8/3} = 11.99 \rightarrow D = 2.53$ m; $B = 1.53$ m; $A = 13.5$ m^2; $V = 15/13.5 = 1.11$ m/s < 2.0 ✔.
- Provide freeboard 0.75 m; lining thickness 10 cm.

17. Solved Problems

❖Problem 1 – Kennedy's V_0

> $D = 2$ m, $m = 1.0$: $V_0 = 0.55 \times 2^{0.64} = 0.55 \times 1.558 = 0.857$ m/s.

❖Problem 2 – Lacey's

> $Q = 100$ m^3/s, $f = 1.0$: $V = (100/140)^{1/6} = 0.945$ m/s; $A = 105.8$ m^2; $P = 4.75 \times 10 = 47.5$ m; $R = 5 \times 0.893/2 = 2.23$ m; $S = 1/(3340 \times 100^{1/6}) = 1/(3340 \times 2.154) = 1/7195$.

❖Problem 3 – Seepage loss

> Wetted perimeter 20 m, length 5 km, loss 2.5 cumec/10^6 m^2: area = 10^5 m^2 → loss = 0.25 m^3/s.

❖Problem 4 – Saving by lining

> Unlined seepage = 3 cumec/million m^2; lined = 0.5; area = 2×10^6 m^2 → saving = $(3 - 0.5) \times 2 = 5$ m^3/s.

18. Important Questions – Unit IV

- Classify canals; canal alignment (ridge, contour, side-slope); inundation canals.
- Cross-section of an irrigation channel – balancing depth, borrow pit, spoil bank, land width.
- Kennedy's theory: assumptions, V_0 equation, design procedure & drawbacks.
- Lacey's regime theory: equations, design; defects; comparison with Kennedy.
- Water logging – causes, effects, remedies; saline and alkaline soils & reclamation.
- Canal losses; lining – necessity, advantages, disadvantages, types; design of lined canal.

Lacey: $V=(Qf^2/140)^{(1/6)}$; $P=4.75\sqrt{Q}$; $R=5V^2/2f$; $S=f^{(5/3)}/(3340 Q^{(1/6)})$; Kennedy: $V0=0.55$ m $D^{0.64}$

WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – V : DIVERSION HEAD WORKS

1. Introduction – Head Works

Head works = structures constructed at the head (start) of a canal to regulate the supply from a river/ reservoir into the canal system. They raise the river level, divert water into the canal, control silt & floods.

❖Types of head works

- (A) Storage head works: dam/ reservoir built across a river to store monsoon flow and release as needed (Nagarjuna Sagar, Srisailam). Used for non-perennial rivers, hydro-power, flood control.
- (B) Diversion head works: weir or barrage across a perennial river which raises (pounds) the water level so that it flows by gravity into the canal (Dowleswaram barrage on the Godavari, Prakasam barrage on Krishna at Vijayawada).

❖Diversion vs storage head works

Point	Diversion	Storage
Purpose	Raise level, divert flow	Store water for later use
Structure	Weir/ barrage (low height)	Dam (high)
River	Perennial, assured flow	Seasonal flow
Storage	Very little (pond)	Large reservoir
Cost	Low	Very high
Silt	Problem (silting of pond)	Settled in reservoir
Examples	Prakasam, Dowleswaram, Sukkur	Srisailam, Hirakud

2. Weirs and Barrages

Weir = a solid obstruction (raised crest) across a river with fixed crest (no gates, or only a few scouring gates), raising water level by the height of the crest; water flows over the crest in floods and the level is maintained by crest height (upstream pond above crest, with shutters on the top).

Barrage = a diversion structure with a low-crest and large gated openings (bays) with piers across the river; pond level is controlled by gates (raised/ lowered) – no (or small) permanent crest; flood passes through the gates (water level in flood regulated by gate opening).

❖Types of weirs (by section)

- Vertical drop (masonry) weir – vertical downstream face; water falls into a water cushion (on the rear) – for hard rock/ strong foundations.
- Rock-fill weir with sloping aprons – loose rock, sloping upstream & downstream aprons; cheap in boulder rivers (Punjab, rocky beds).
- Concrete weir with (sloping) glacis – modern – concrete floor with glacis (slope 3:1 to 2:1), hydraulic jump forms on glacis/ at the toe dissipating energy: Type A (jump on the floor), Type B (jump on the glacis), Type C (jump at downstream of glacis) – adopted on permeable (alluvial) foundations – most common.
- Cut-off wall / sheet-piles, impervious floor provide seepage safety.
- Drum/ rubber (inflatable) weirs; shutter weirs.

❖Weir vs barrage

Feature	Weir	Barrage
Crest	High, fixed crest	Low crest, gates control level
Pond control	Crest height + shutters	Gates
Flood passage	Over crest (afflux ↑)	Through gate openings (less afflux)
Cost	Lower (less steel/ gates)	Higher (gates, piers, bridge)
Silt control	Less	Better (gates flush silt)
Afflux & submergence	Higher	Lower
Road bridge	Not usually	Often combined (traffic bridge over piers)
Examples	Anicut (Tamil Nadu), Okhla (old)	Sukkur, Farakka, Prakasam, Dowleswaram, Dhavaleswaram

3. Layout of Diversion Head Works & Components

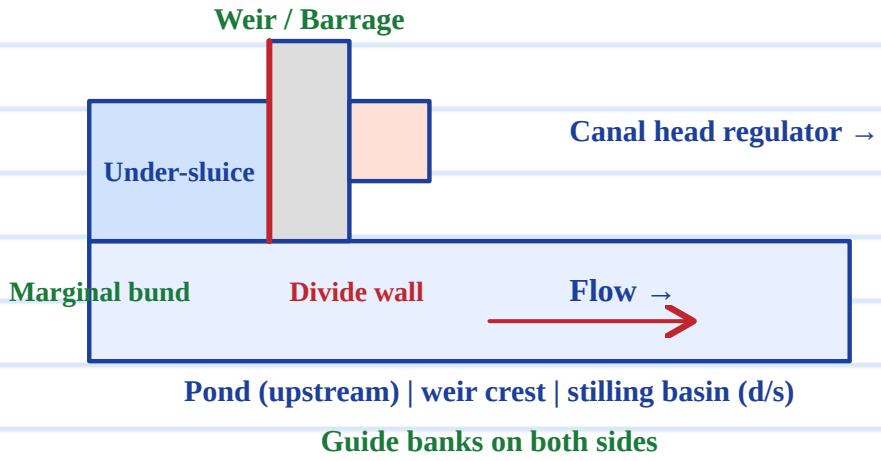


Fig 5.1 Layout of diversion head works (plan, simplified)

❖General layout (from upstream to downstream, plan)

- River → upstream marginal bunds & guide banks → weir/ barrage (main river section) + under-sluices (scouring sluices) on the canal side → divide wall → canal head regulator at the off-take, silt excluder in front of regulator, fish ladder, river training works, downstream (stilling basin/ energy dissipation), cut-offs & aprons.

❖Components

- 1. Weir/ barrage proper: raises the pond level; crest & gates; impervious floor (concrete), glacis, piers.
- 2. Under-sluices (scouring sluices): gated openings at the canal end of the weir (on the canal side), with crest level lower than the weir crest (usually at river bed level); purposes: (i) scour the silt deposited in front of the canal head regulator, (ii) maintain a clear, deep channel towards the head regulator, (iii) pass low floods without opening weir gates, (iv) regulate the river.
- 3. Divide wall: masonry/ concrete wall (RC) separating the under-sluice portion from the weir and providing a still pocket for silt excluders; perpendicular to the weir axis; extends up to the beginning of the upstream pond & beyond the downstream floor; prevents cross-flow from the weir into the under-sluice pocket; cross-currents are controlled.
- 4. Fish ladder: a passage (step-pool) for fish migration across the weir – a channel with baffles/ steps (1 in 10 slope), near divide wall.

- 5. Canal head regulator: gated structure at the head of the canal controlling the discharge entering the canal (also prevents entry of silt & flood water); crest level ≥ 1.2 m higher than under-sluice crest; aligned at 90° – 110° to the weir axis (to reduce silt entry).
- 6. Silt control devices: Silt excluders (in front of canal head – separate silt-laden bottom layer & divert it into the under-sluices; tunnels under head regulator bay) and silt ejectors (extractors) (on the canal – just beyond the head regulator – remove the silt that has already entered the canal; tunnel type).
- 7. River training works: guide banks (guide the river flow axial to the weir, trumpet shaped, length ≥ 1.0–1.5 × width of waterway, prevent river from outflanking), marginal (afflux) bunds (protect the upstream land/ banks from submergence due to afflux; each side, freeboard 1.5 m above HFL + afflux), spurs/ groynes (to deflect flow, protect banks).
- 8. Pond/ Upstream pocket: the pool (pondage) upstream of the weir; pond level = crest level + shutters or gate level; afflux = rise in HFL due to the weir.
- 9. Floor, aprons, cut-offs (sheet piles), protection works: upstream & downstream impervious floor, stilling basin, toe walls/ cut-off, inverted filter, launching apron (loose stone/ cc blocks) on both ends, upstream blocks.
- 10. Trash rack, stop-log gates, service bridge/ gate-operating deck.

❖Location of head works

- Straight, stable reach of river, with good rock/ clay foundation, high banks & minimum cross-drainage; narrow river width; above the command area (commanding level); near the canal alignment; good approach for construction.

4. Failure of Hydraulic Structures on Permeable Foundations – Causes

> On pervious (sandy/ alluvial) foundations, the weir is subject to seepage (water percolating under the structure) from the upstream pond to downstream; this gives uplift pressure and exit gradients, causing failures.

❖A. Failure due to subsurface flow

- (i) Piping (undermining): seepage water emerges at the downstream end with a velocity (hydraulic gradient > critical), carrying away fine soil grains ("boiling/ quicksand" condition), forming a pipe-like channel ("pipe") backward under the floor → progressive undermining, floor settles, collapse.
  - Critical (exit) gradient $i_c = (G_s - 1)(1 - n) \approx 1.0$ ($G_s = 2.65$, $n = 0.4 \rightarrow 0.99$); safe gradient is 1/5 – 1/7.
- (ii) Rupture of floor by uplift pressure: uplift (upward water pressure) on the underside of the floor is more than the weight of floor + water above → floor cracks/ bursts (uplift is highest at upstream end of floor) → seepage increases → failure.

❖B. Failure due to surface flow

- (iii) Scour/ erosion: high-velocity flow from the structure scours the river bed & banks downstream (retrogression) & upstream of the pier ends, undermining the floor ends.
- (iv) Suction caused by standing wave (hydraulic jump): negative pressure leads to removal of floor blocks.
- (v) Rupture by hydraulic jump (impact) if the jump forms on the floor without enough thickness; excess velocity damages the floor/ stilling basin.
- (vi) Failure by overturning/ sliding of the weir body (not due to seepage – instability).
- (vii) Impact of floating debris, silting, downstream retrogression of bed level.

❖Remedies (safety measures)

- Provide upstream & downstream cut-offs (sheet piles/ concrete) – increase the creep path & reduce exit gradient, cut-off at downstream (deeper) to protect against piping.
- Provide a longer impervious floor/ blanket; proper floor thickness (against uplift); inverted filter + pervious floor (weep holes) with launching aprons on the downstream end (to relieve uplift & stop piping).
- Stilling basin with baffle blocks; proper energy dissipation; protect toe with CC blocks & loose stone aprons; pressure-relief drainage holes/ pipes.
- Provide "pervious floor" downstream & "impervious floor" upstream.

5. Seepage Theories – Design of Impervious Floor

- Seepage under the weir follows curved path (streamlines) from upstream cut-off/ floor to the downstream exit; the head H (upstream pond level – downstream tail water level) is lost along the path.

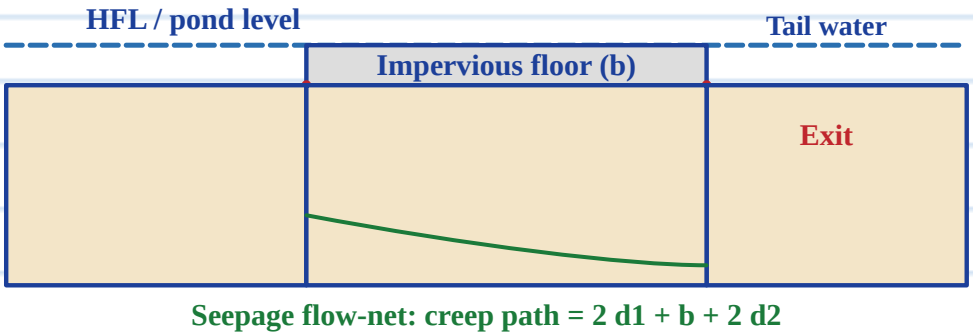


Fig 5.2 Impervious floor with upstream & downstream cut-off piles (seepage path)

- Design requirements: (i) safe exit gradient (no piping); (ii) floor thickness safe against uplift. Floor length & cut-off depths are fixed by seepage theories (Bligh – Lane – Khosla).
- Theories: Bligh's creep theory (1910), Lane's weighted creep, Khosla's method (1936 – flow net/ mathematical solutions – scientifically most accurate).

6. Bligh's Creep Theory

> Bligh: the water seeping under a weir creeps along the contact of the floor and the foundation (along the surface of the structure); the head loss is proportional to the length of creep (percolating water loses head at a constant rate along its path). Total creep length $L = L_h + L_v$ (horizontal + vertical, with no distinction).

- Creep length $L = b + 2\,d_1 + 2\,d_2 + \ldots$ = total (horizontal floor length b + twice the depth of each pile/ cut-off (down & up)). (Slopes steeper than 45° counted as vertical; flatter counted as horizontal.)

❖Assumptions

- Loss of head ∝ creep length (uniform hydraulic gradient H/L); vertical and horizontal creep are equally effective; the water particle follows the structure surface; no effect of pile positions/ concentrations; no change of gradient with depth.

❖Safety against piping

$$L = C\,H \rightarrow \text{hydraulic gradient } H/L = 1/C \text{ (must be} \leq 1/C\text{)}$$

- C = Bligh's coefficient (depends on soil; finer soil → larger C → longer creep path needed).

Type of soil	Bligh's C	Safe gradient 1/C
Light sand & mud	18	1/18
Fine micaceous sand	15	1/15
Coarse grained sand	12	1/12
Gravel & sand (boulders)	9	1/9
Boulders, shingle, gravel & sand	4 – 9	1/4 – 1/9

- Rule: provided creep length $\geq C\,H$.

❖Safety against uplift (floor thickness)

- Residual (uplift) head at a point at creep distance l from the upstream end:

$$h_l = H - (H/L)\cdot l \text{ (m of water)}$$

- Uplift pressure $p = \gamma_w h_l$ (acting on the underside); the floor must resist this by its own weight:

$$t = (4/3)\cdot h/(G-1),\ G = \text{specific gravity of floor material } (\approx 2.24 \text{ for concrete})$$

- Derivation: $(t + h_{top})\,\gamma_w$ ... weight of floor per unit area in water = $(G-1)\,\gamma_w\,t \geq \gamma_w\,h \rightarrow t = h/(G-1)$; extra 1/3 (safety) → $t = (4/3)\,h/(G-1)$. For $G = 2.24$: $t \approx 1.075\,h$.
- Floor must be thick enough at the points where h is greatest (upstream end); thickness increases from the downstream end towards where uplift is greater.

❖Limitations / defects of Bligh's theory

- Does not distinguish between horizontal & vertical creep (vertical piles are more effective!); no distinction between outer & inner faces of piles.
- Neglects position of cut-offs (a downstream pile is more effective than the same pile upstream); uniform hydraulic gradient (loss of head ∝ length) is wrong – actual loss is non-uniform (rapid near ends).
- Does not check safety against exit gradient (the true cause of piping) – only a "safe" average gradient; no uplift pressure variations/ mutual interference of piles considered.
- Provides approximate safe results for small works only; overdesigned or unsafe for pile at the end.
- Lane's weighted creep theory (improvement): $L_w = \Sigma L_v + (1/3)\Sigma L_h \geq C_w H$; C_w = 8.5 (very fine sand/ silt), 7.0 (fine sand), 6.0 (medium sand), 5.0 (coarse sand), 4.0 (fine gravel), 3.0 (coarse gravel/ soft clay), 2.5 (boulders), 1.8 (hard clay).

❖Example – Bligh

> A weir has floor 40 m long with upstream cut-off 3 m and downstream cut-off 4 m deep; head H = 4.5 m; soil: coarse sand C = 12.

- Creep length L = 40 + 2×3 + 2×4 = 54 m; required = CH = 12 × 4.5 = 54 m ✔ (just safe).
- Uplift at point A (d/s end of floor, before the d/s pile): creep up to A = 3 + 3 + 40 = 46 m → h = 4.5 – (4.5/54)(46) = 0.667 m; t = (4/3)(0.667/1.24) = 0.72 m.
- Uplift at toe of u/s pile: l = 3 → h = 4.5 – 0.25 = 4.25 m.

7. Khosla's Theory

> Khosla (1936): Seepage under a structure occurs in curved stream-lines and equipotential lines forming a flow net; the flow is governed by Laplace's equation ($\nabla^2\varphi = 0$). Khosla & co-workers solved the equation (Joukowski/ conformal transformation) for simple standard profiles and gave the pressure at key points and the exit gradient. Complex profiles are broken into simple standard forms. Results are expressed as percentage of head (φ).

- Key points: the junctions of the floor with pile lines and the bottom of piles (points E, D, C) in simple profiles; upstream floor start = 100%, downstream tail = 0%.
- Concept: loss of head isn't proportional to the creep length; piles (cut-offs) are much more effective at the ends; downstream pile controls the exit gradient, upstream pile reduces the uplift on the floor.

❖Standard profiles (Khosla's simple forms)

- (a) Straight floor with a pile at the upstream end (U/S pile).
- (b) Straight floor with a pile at the downstream end (D/S pile).
- (c) Floor with an intermediate pile.
- (d) Floor depressed/ raised (sloping), floor with two piles → solved by combining & correcting these.

❖Notation

- b = total floor length (horizontal); d = depth of the pile; $\alpha = b/d$; $\lambda = [\,1 + \sqrt{(1+\alpha^2)}\,]/2$; H = maximum seepage head; φ = pressure expressed as % of H (residual).

❖Pressure percentages (pile at the downstream end, i.e., b/d profile)

$$\varphi_E = (1/\pi)\cos^{-1}[(\lambda-2)/\lambda]\quad \varphi_D = (1/\pi)\cos^{-1}[(\lambda-1)/\lambda]\quad \varphi_C = 0$$

- (angle in radians; result × 100 gives %; i.e., $\varphi(\%) = \cos^{-1}(\ldots)$(degrees) ÷ 1.8)

❖Pile at the upstream end

$$\varphi_{E1} = 100\%\quad \varphi_{D1} = 100 - \varphi_D\quad \varphi_{C1} = 100 - \varphi_E$$

- (use the same φ_E, φ_D expressions with $\alpha = b/d$ for the u/s pile).
- Intermediate pile: pressures at the points E, D, C of the pile are read from Khosla's curves (depending on $\alpha = b/d$ and the position of pile b_1/b), or by the same type of expressions with modified α.

❖Exit gradient

> Exit gradient G_E = hydraulic gradient of the seepage at the downstream end of the floor (exit), the upward gradient at the toe; if it exceeds the critical value (≈ 1) → piping.

$G_E = (H/d) \cdot [1/(\pi\sqrt{\lambda})]$ (d = depth of downstream cut-off)

- With λ as above ($\alpha = b/d$). For a long floor (large α): $G_E \approx 1.41\, H/(\pi\sqrt{b\,d})$.
- Safe exit gradients (Khosla):

Soil	Safe G_E
Shingle (gravel)	1/4 – 1/5
Coarse sand	1/5 – 1/6
Fine sand	1/6 – 1/7
Critical gradient (all)	≈ 1 ($i_c = (G_s - 1)(1 - n)$)

- Condition: $G_E \leq$ safe value → increase d (deeper d/s cut-off), or b, to reduce G_E. Factor of safety 4–6.
- Exit gradient charts: G_E d/H plotted vs $1/\alpha$.

8. Corrections to Khosla's Pressures

- Since the actual profile is a combination of piles and floor, the pressures read for a simple profile need correction:
- (1) Correction for mutual interference of piles: piles influence each other; correction $C\,(\%) = 19\sqrt{(D/b')} \times (d + D)/b$
  - D = depth of the influencing pile line (the pile whose effect on another pile is being found); d = depth of the pile on which the influence is calculated; b' = distance between the two piles; b = total floor length.
  - Sign: positive (+) for points in the rear (behind, i.e. upstream side) of the influencing pile; negative (–) for points ahead of it in the direction of flow.
- (2) Correction for floor thickness: key points E, C are at the top surface of the floor in Khosla's curves; actual pressure is needed at the bottom of the floor; interpolate linearly between the pile-bottom and floor-top values: (E' = pressure at the bottom of floor) corrected: E: negative, C: positive correction ($\Delta\Phi = (\varphi_D - \varphi_E)\, t/d$).
- (3) Correction for slope of floor: $C_s = C \times b_s / b_1$ (Khosla's table of C for slope 1:1…1:6 – positive for downward slope (in the direction of flow), negative for upward slope; applies to the pressure at the key point at the start/ end of the slope).

❖Steps in design by Khosla's method

- 1. Fix H, soil type & safe exit gradient; choose d/s cut-off depth d (below scour level, ≥ 1.2–1.5 × scour depth) and floor length b (initially from Bligh/ Lane).
- 2. Compute $\alpha = b/d$, λ; $G_E = H/(d\pi\sqrt{\lambda})$; compare with safe value; modify d or b if needed.
- 3. Find φ_E, φ_D, φ_C for each pile (standard form); apply corrections (interference, thickness, slope).
- 4. Residual uplift head at each key point = (φ%/100) × H – (head above floor/ hydrostatic); thickness of floor $t = h/(G - 1)$ (or $4/3 \times h/(G-1)$); fix t at key points; provide a nominal minimum (0.5–0.6 m) under the crest**.
- 5. Check overturning/ sliding; design aprons, filters.

❖Example – Khosla (D/S pile)

> Floor b = 30 m, d/s pile d = 3 m, H = 5 m. $\alpha = 30/3 = 10$; $\lambda = [1 + \sqrt{(1 + 100)}]/2 = (1 + 10.05)/2 = 5.525$.

- $\varphi_E = (1/\pi)\cos^{-1}[(5.525 - 2)/5.525] = \cos^{-1}(0.638)/180° = 50.4/180 = 28.0$ % → pressure head = 0.28 × 5 = 1.40 m.
- $\varphi_D = \cos^{-1}[(5.525 - 1)/5.525 = 0.819]/180 = 35.0/180 = 19.4$ % → 0.97 m.
- Exit gradient $G_E = (5/3) \times 1/(\pi \times \sqrt{5.525}) = 1.667 \times 0.1354 = 0.226 \approx 1/4.4$ → > safe 1/6 (fine sand) ✗ → increase d to ≈ 5 m: $\alpha = 6$, $\lambda = (1+\sqrt{37})/2 = 3.54$; $G_E = (5/5) \times 1/(\pi \times 1.88) = 0.169 \approx 1/5.9$ ✔ (safe for coarse sand).

9. Determination of Uplift Pressure & Impervious Floor Length

❖By Bligh

- Total creep $L = C\,H$ gives minimum floor length: $b = C\,H - 2\Sigma\, d$ (with cut-offs).
- Uplift head at any point: $h = H - (H/L)\,l$; floor thickness $t = (4/3)\,h/(G - 1)$ (with G = 2.24).
- Weak point of Bligh: only the average; may be unsafe against piping because exit gradient is not checked.

❖By Khosla

- Uplift at key points obtained from φ values (after corrections); residual head at floor point = $\varphi \times H$; net uplift = $h_{residual}$ – (hydrostatic head above floor point); thickness $t = h_{net}/(G - 1)$; exit gradient G_E check.
- More realistic & recommended; flow net verified by electrical/ sand analogy and numerical (FEM).

❖Comparison

Point	Bligh	Khosla
Basis	Empirical, creep length	Flow net/ potential theory (Laplace)
Head loss	Uniform along creep length	Non-uniform, depends on profile
Vertical vs horizontal creep	Equal	Different effectiveness
Pile position/ depth	Ignored	Accounted (u/s pile reduces uplift; d/s pile reduces exit gradient)
Exit gradient	Not checked	Calculated & checked
Uplift pressure	Linear variation	Accurate (curved) distribution
Corrections	None	Interference, thickness, slope
Accuracy	Approximate	High – adopted for design
Use	Preliminary/ small works	Final design

10. Solved Problems

❖Problem 1 – Bligh

Fine sand $C = 15$, $H = 3$ m: creep length required $L = 45$ m; with piles 2 m + 3 m → floor $b = 45 - 2(2+3) = 35$ m.

❖Problem 2 – Exit gradient

$H = 4$ m, $d = 2.5$ m, $b = 25$ m: $\alpha = 10$, $\lambda = 5.525$ → $G_E = (4/2.5)/(\pi \times 2.35) = 0.217$ ($> 1/6$ ✗). With $d = 4$ m, $\alpha = 6.25$, $\lambda = (1+\sqrt{40.06})/2 = 3.66$ → $G_E = 1/(\pi \times 1.913) = 0.166$ ($\approx 1/6$) ✔.

❖Problem 3 – Uplift & thickness (Khosla)

φ_E = 28% at E, H = 5 m → residual pressure head = 1.4 m. Taking the net uplift = 1.4 m (conservative, ignoring water cover on the floor): t = 1.4/(2.24 – 1) = 1.13 m; with the 4/3 safety factor t ≈ 1.5 m.

❖Problem 4 – Floor thickness by Bligh

Residual head at a point = 2.5 m, G = 2.24 → t = (4/3)(2.5/1.24) = 2.69 m.

❖Problem 5 – Critical gradient

G_s = 2.65, n = 0.40 → i_c = 1.65 × 0.6 = 0.99 ≈ 1; safe with FOS 5 = 0.2 = 1/5.

II. Important Questions – Unit V

- Types of diversion head works; differentiate diversion and storage; weir vs barrage.
- With a neat layout, explain the components of diversion head works (under-sluices, divide wall, fish ladder, head regulator, silt excluder/ ejector, guide banks).
- Causes of failure of hydraulic structures on permeable foundations & remedies.
- Bligh's creep theory: assumptions, design for piping & uplift; limitations.
- Khosla's theory: standard profiles, pressures φ_E, φ_D, φ_C, corrections; exit gradient; safe values.
- Compare Bligh's & Khosla's theories.

Key formulas: $L = C\,H$; $t = (4/3)\,h/(G-1)$; $\lambda = [1+\sqrt{(1+\alpha^2)}]/2$; $G_E = H/(d\pi\sqrt{\lambda})$; Safe G_E: 1/5 – 1/7